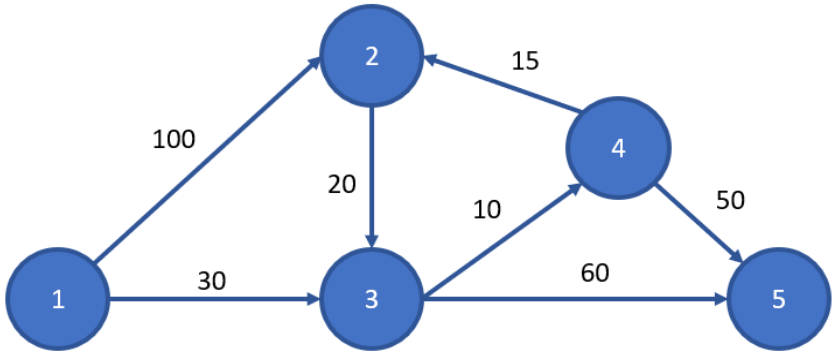
# Networks

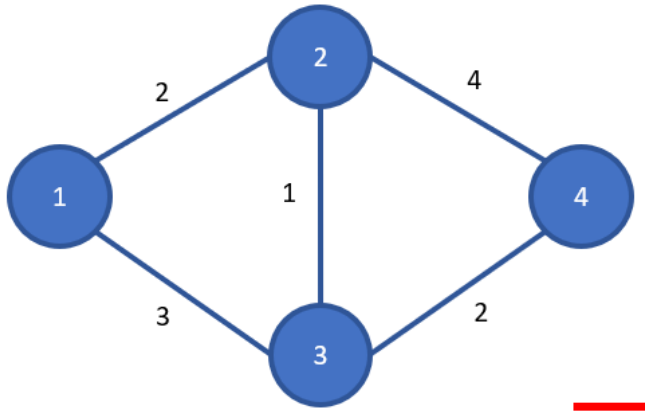
## Shortest Paths and Minimum Spanning Tree (MST)

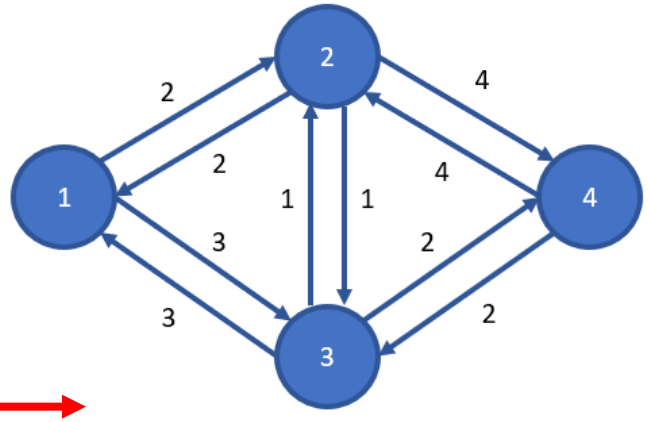
Sensitivity: Internal

# Network

Directed graph $G = (V, E)$, where $E$ is a set of edges, subset of $\mathrm{V} \times \mathrm{V}$, and a weight function $w$: $E \to \mathbb{R}$




$V = \{1,2,3,4,5\}$
$\mathrm{E} = \{(1,2), (1,3), (2,3), (3,4), (3,5), (4,2), (4,5)\}$

e.g. $w(1,2) = 100$

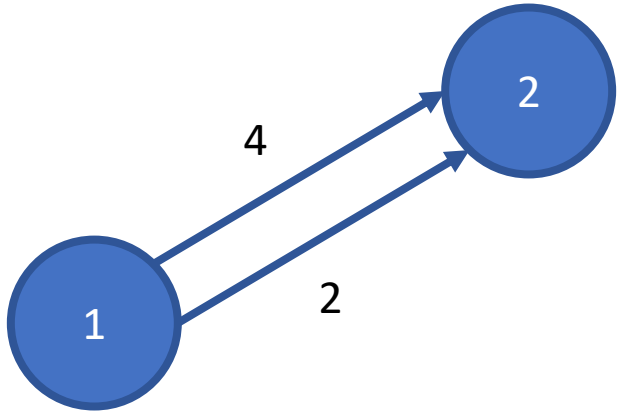
$V = \{1,2,3,4\}$
$E = \{(1,2), (1,3), (2,3), (2,4), (3,4)\}$

$V = \{1,2,3,4\}$
$E = \{(1,2), (1,3), (2,1), (2,3), (2,4), (3,1), (3,2), (3,4), (4,2), (4,3)\}$

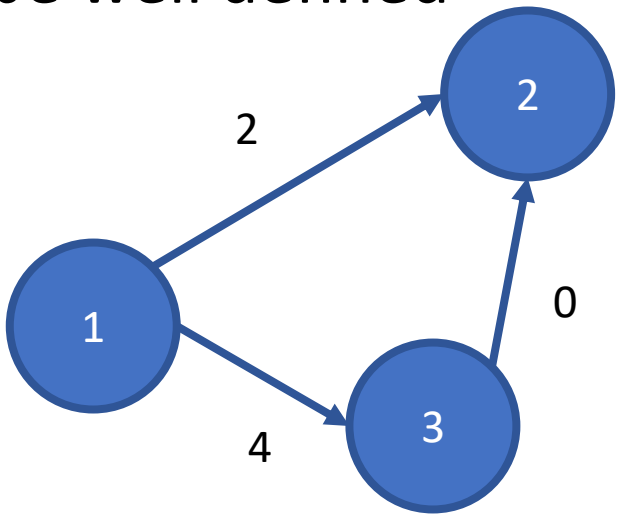
Sensitivity: Internal

# Network

Note we do not allow multiple edges between vertices $u$ and $v$, otherwise the edge weight would not be well defined




What is $w(1,2)$?




What is $w(1,2)$?

$$w(1,2) = 2$$

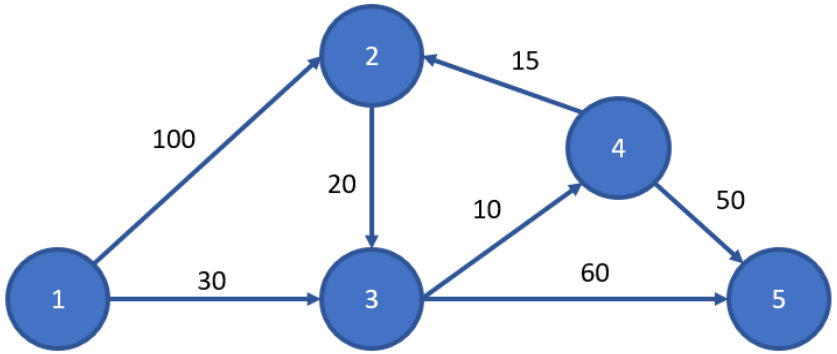
Alternative path from 1-2 is 1-3-2

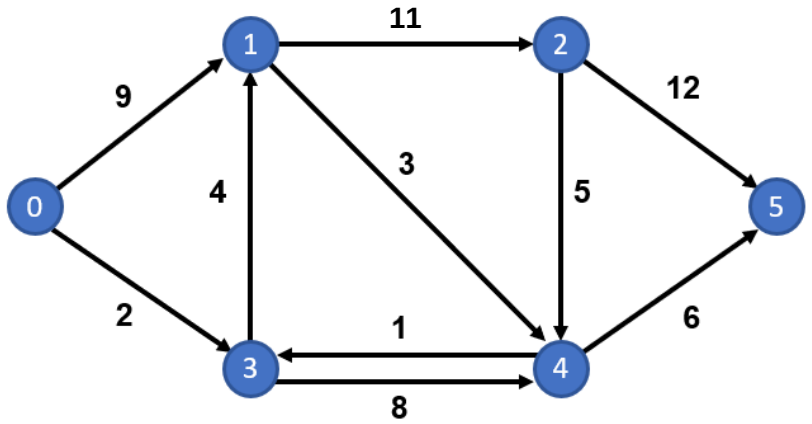
$$w(1,3) + w(3,2) = 4 + 0 = 4$$

Sensitivity: Internal

# Adjacency Matrix

We won't have direct use for this, but it is worth noting!




$$\begin{pmatrix} 0 & 100 & 30 & - & - \\ - & 0 & 20 & - & - \\ - & - & 0 & 10 & 60 \\ - & 15 & - & 0 & 50 \\ - & - & - & - & 0 \end{pmatrix} \textbf{ OR } \begin{pmatrix} 0 & 100 & 30 & \infty & \infty \\ \infty & 0 & 20 & \infty & \infty \\ \infty & \infty & 0 & 10 & 60 \\ \infty & 15 & \infty & 0 & 50 \\ \infty & \infty & \infty & \infty & 0 \end{pmatrix}$$

$$\begin{pmatrix} 0 & 9 & - & 2 & - & - \\ - & 0 & 11 & - & 3 & - \\ - & - & 0 & - & 5 & 12 \\ - & 4 & - & 0 & 8 & - \\ - & - & - & 1 & 0 & 6 \\ - & - & - & - & - & 0 \end{pmatrix} \textbf{ OR } \begin{pmatrix} 0 & 9 & \infty & 2 & \infty & \infty \\ \infty & 0 & 11 & \infty & 3 & \infty \\ \infty & \infty & 0 & \infty & 5 & 12 \\ \infty & 4 & \infty & 0 & 8 & \infty \\ \infty & \infty & \infty & 1 & 0 & 6 \\ \infty & \infty & \infty & \infty & \infty & 0 \end{pmatrix}$$

Sensitivity: Internal

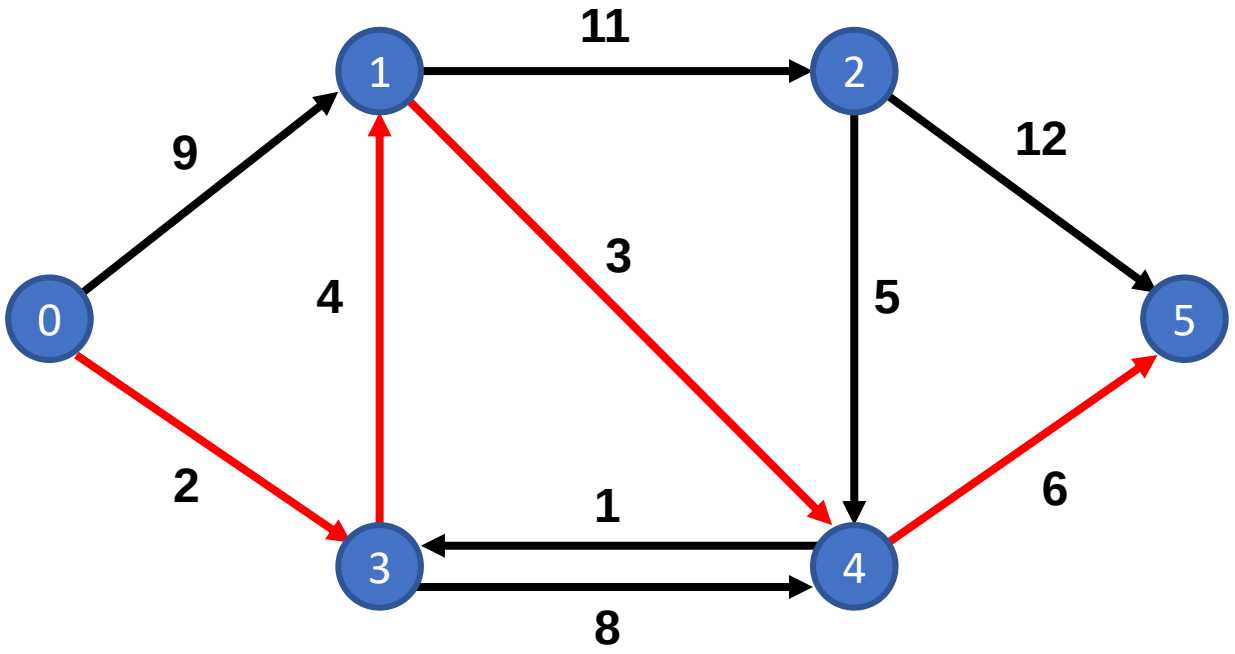
# Shortest Path Problem

**Definition:** Find the shortest path between two vertices (source and destination) such that the sum of the weights of its edges is minimised.

- Perhaps one of the most fundamental problems in Mathematics and Computer Science
- Huge implications on routing and traffic problems, e.g. vehicular, telecommunications.
- If you've used Google Maps, you've asked to solve a form of shortest path problem!

Sensitivity: Internal

# Blue Peter Example




Shortest path between 0 and 5 is 0-3-1-4-5 of length 15

Sensitivity: Internal

# Dijkstra's Algorithm

- Designed by Edsger Wybe Dijkstra in 1956 (published 1959) to demonstrate the new ARMAC computer
- Still widely used and researched today!
- Route planners use pre-processing speed-up techniques such as contraction hierarchies.

Edge weights need to be nonnegative! Otherwise use Bellman-Ford

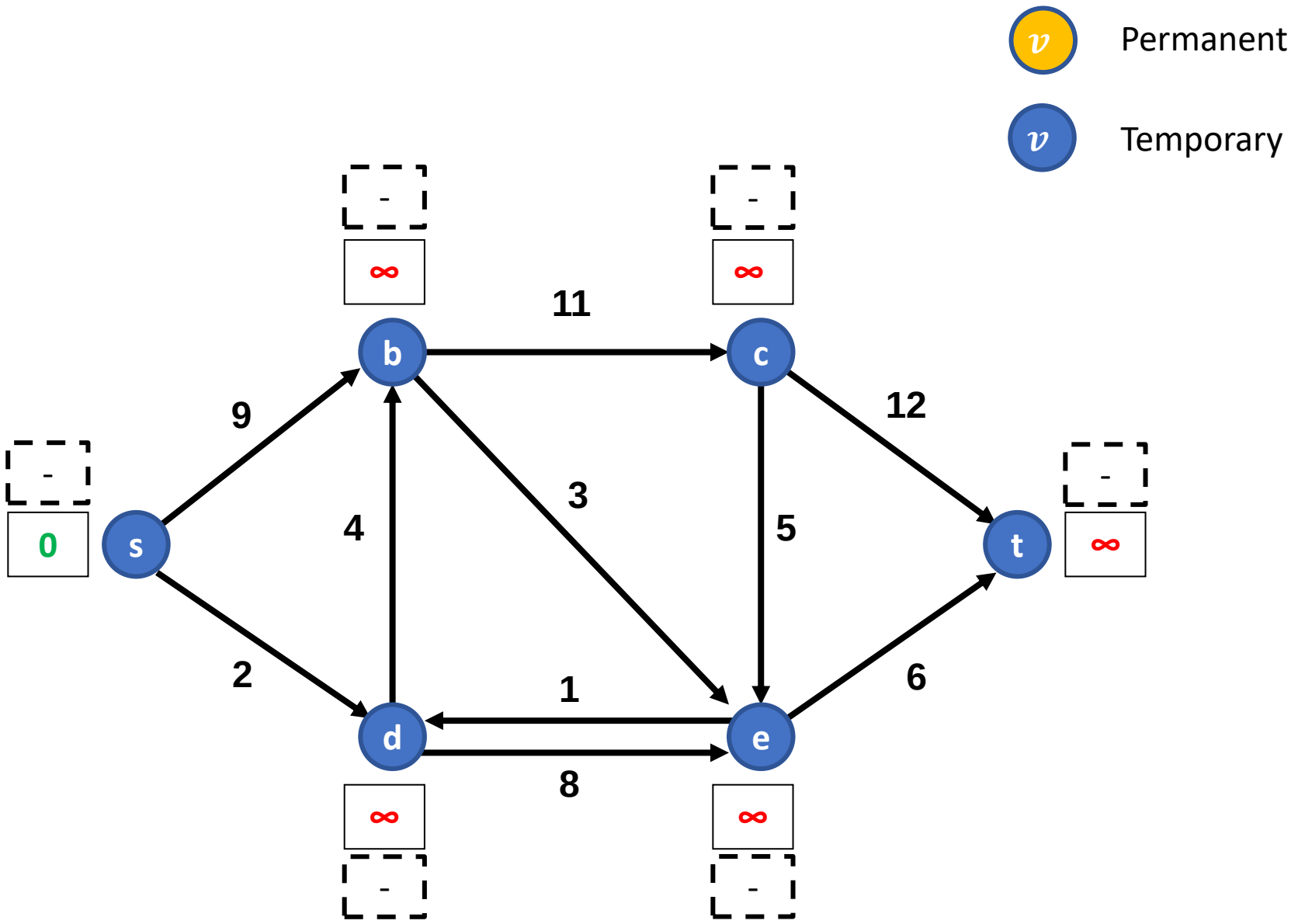
Homework: Investigate Bellman-Ford Algorithm, this can also detect negative weight cycles

1. Initialise the network, source vertex has a distance of 0 and all other vertices a distance of ∞, label all vertices as **temporary**
2. Label the vertex with minimum distance as **permanent**
3. Update the distances and predecessors to reachable **temporary** vertices from the current permanent vertex
4. Repeat steps 2 and 3 until all vertices are marked **permanent**

Sensitivity: Internal

# Set Up

| $v$ | Label | $d_v$ | $\pi_v$ |
|---|---|---|---|
| s | T | 0 | - |
| b | T | | |
| c | T | | |
| d | T | | |
| e | T | | |
| t | T | | |



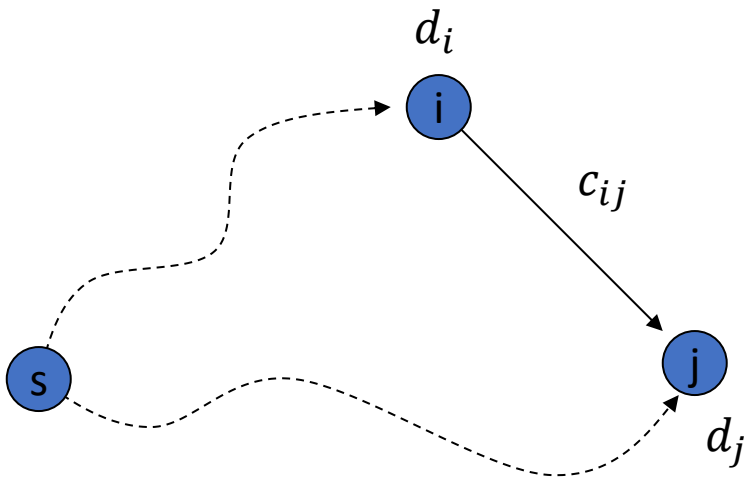

Initialise the network, source vertex has a distance of 0 and all other vertices a distance of ∞

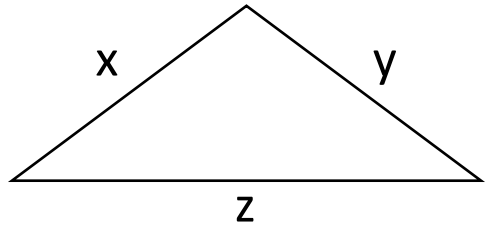
Sensitivity: Internal

# Improving Cost Labels

- Shortest path costs satisfy the triangle inequality:

  $d_i + c_{ij} \geq d_j$ for every edge $(i, j)$.



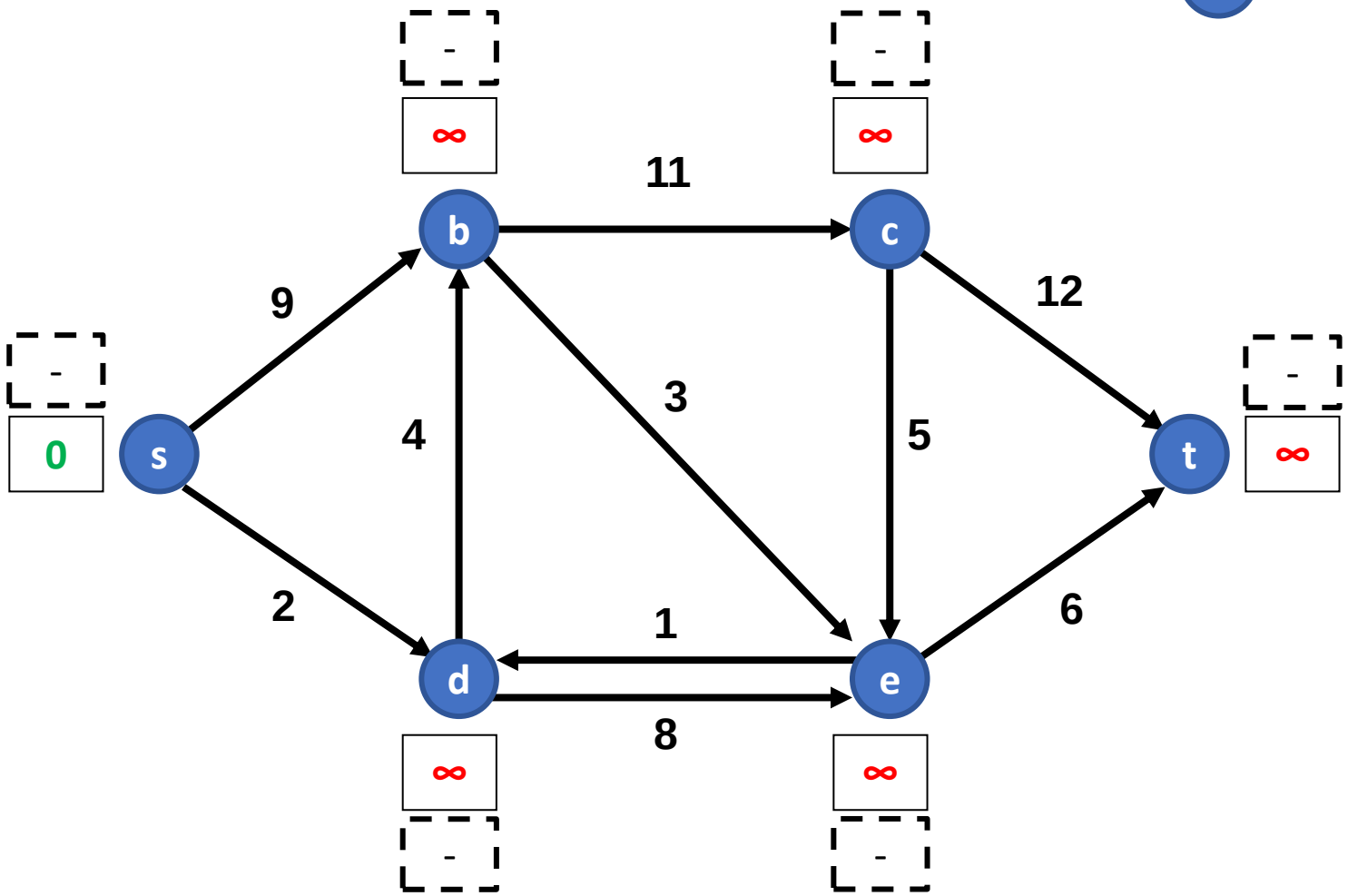

x + y ≥ z

- If this is not **TRUE** then there exists some path that improves $d_j$
  - E.g. The path to $d_i$ + the edge $(i, j)$

Sensitivity: Internal

# Dijkstra Example

| $v$ | Label | $d_v$ | $\pi_v$ |
|---|---|---|---|
| s | T | 0 | - |
| b | T | | |
| c | T | | |
| d | T | | |
| e | T | | |
| t | T | | |

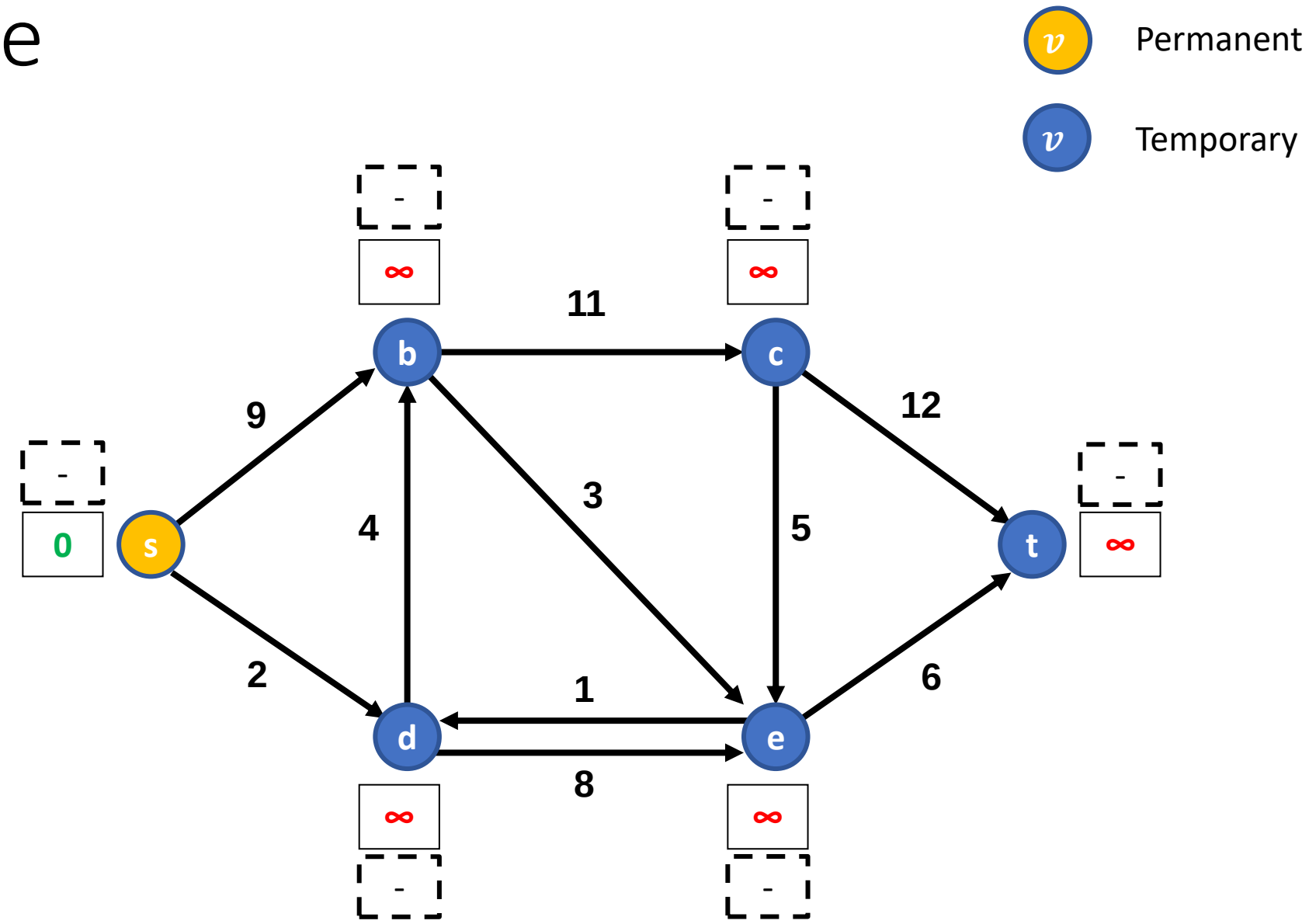



Initialise the network, source vertex has a distance of 0 and all other vertices a distance of ∞

Sensitivity: Internal

# Example

| $v$ | Label | $d_v$ | $\pi_v$ |
|---|---|---|---|
| s | P | 0 | - |
| b | T | | |
| c | T | | |
| d | T | | |
| e | T | | |
| t | T | | |

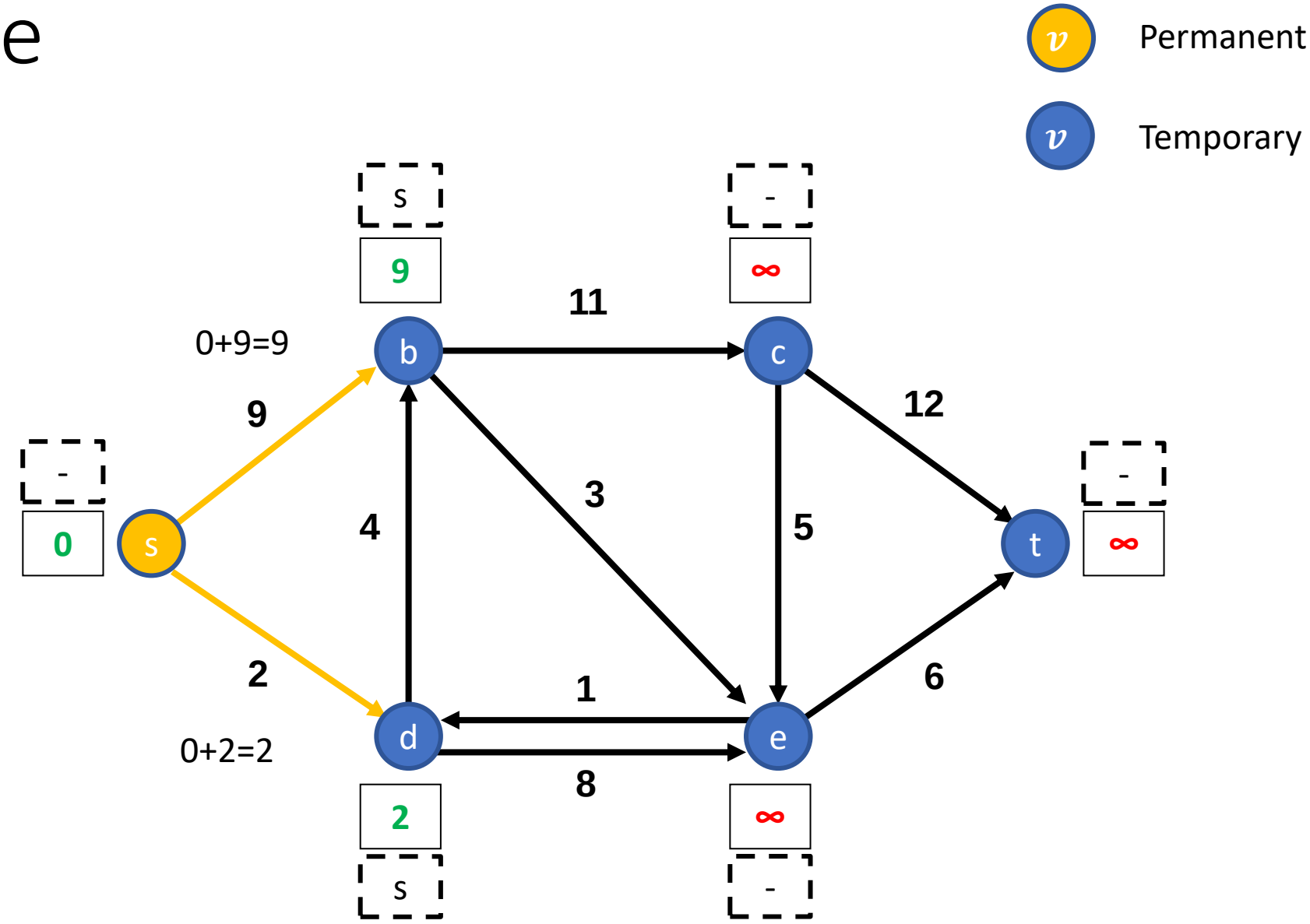



Label temporary vertex with minimum distance as permanent

Sensitivity: Internal

# Example

| $v$ | Label | $d_v$ | $\pi_v$ |
|---|---|---|---|
| s | P | 0 | - |
| b | T | 9 | s |
| c | T | | |
| d | T | 2 | s |
| e | T | | |
| t | T | | |

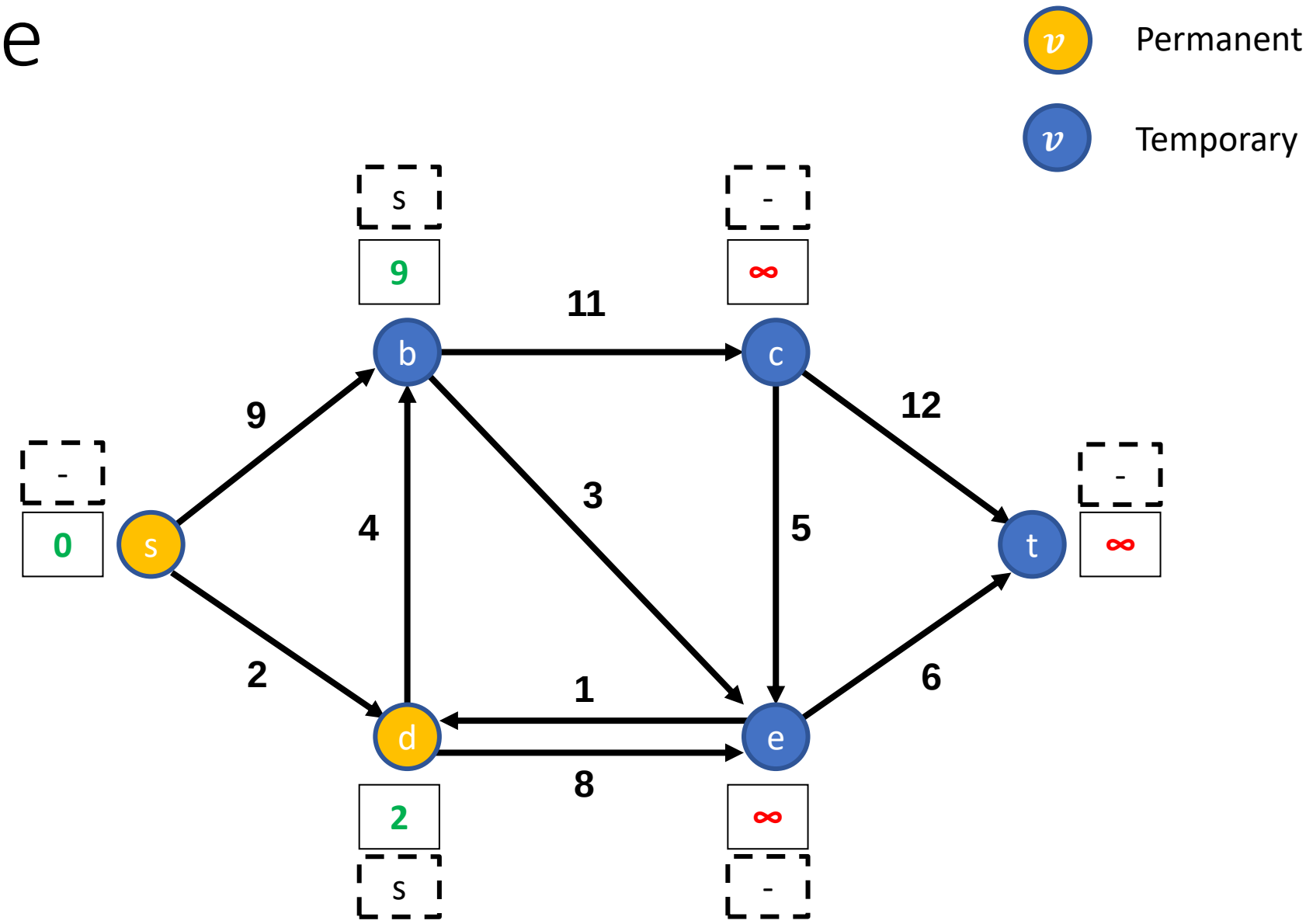



Update distances to reachable vertices which are labelled temporary

Sensitivity: Internal

# Example

| $v$ | Label | $d_v$ | $\pi_v$ |
|---|---|---|---|
| s | P | 0 | - |
| b | T | 9 | s |
| c | T | | |
| d | P | 2 | s |
| e | T | | |
| t | T | | |

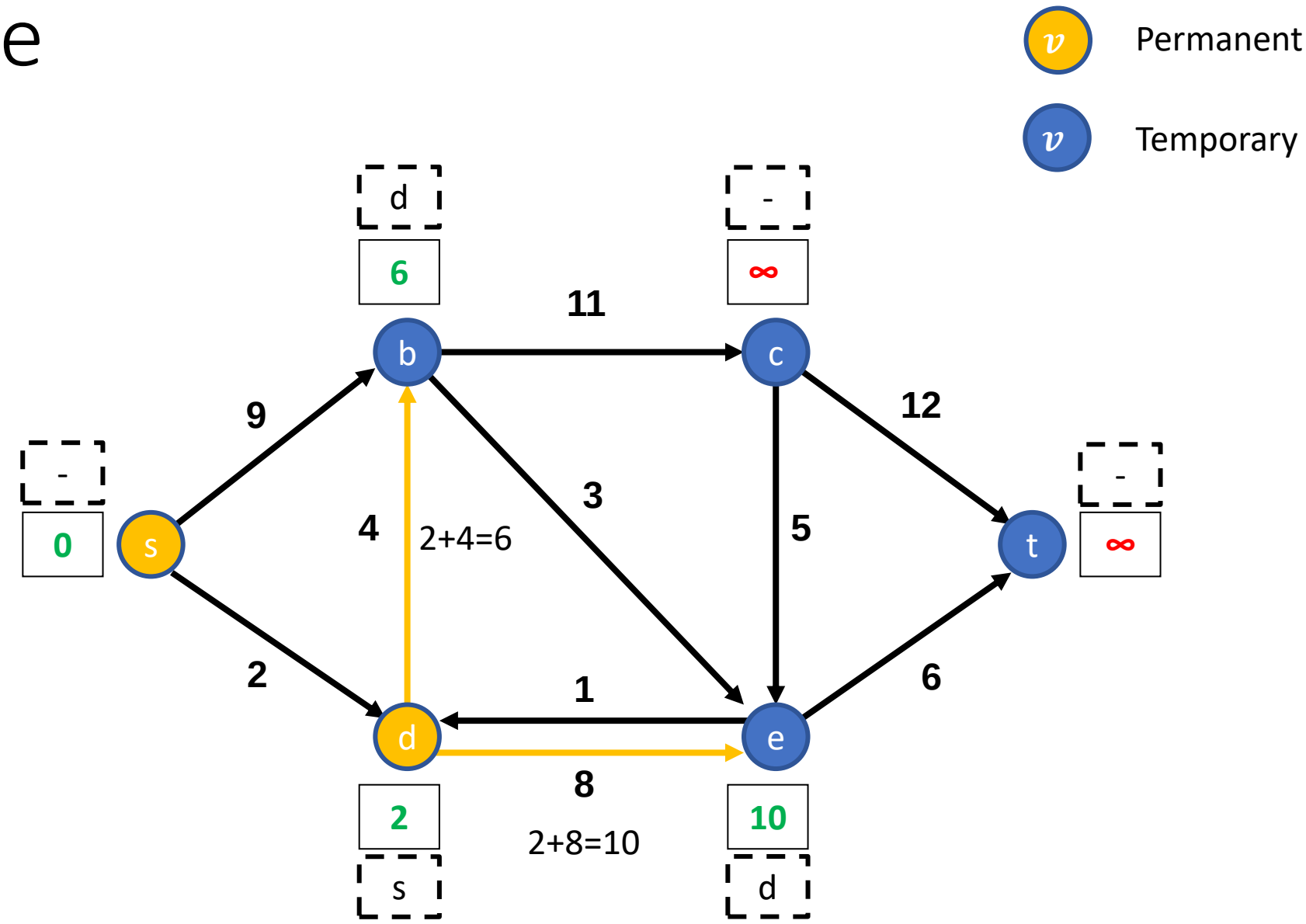



Label temporary vertex with minimum distance as permanent

Sensitivity: Internal

# Example

| $v$ | Label | $d_v$ | $\pi_v$ |
|---|---|---|---|
| s | P | 0 | - |
| b | T | ~~9~~ 6 | ~~s~~ d |
| c | T | | |
| d | P | 2 | s |
| e | T | 10 | d |
| t | T | | |

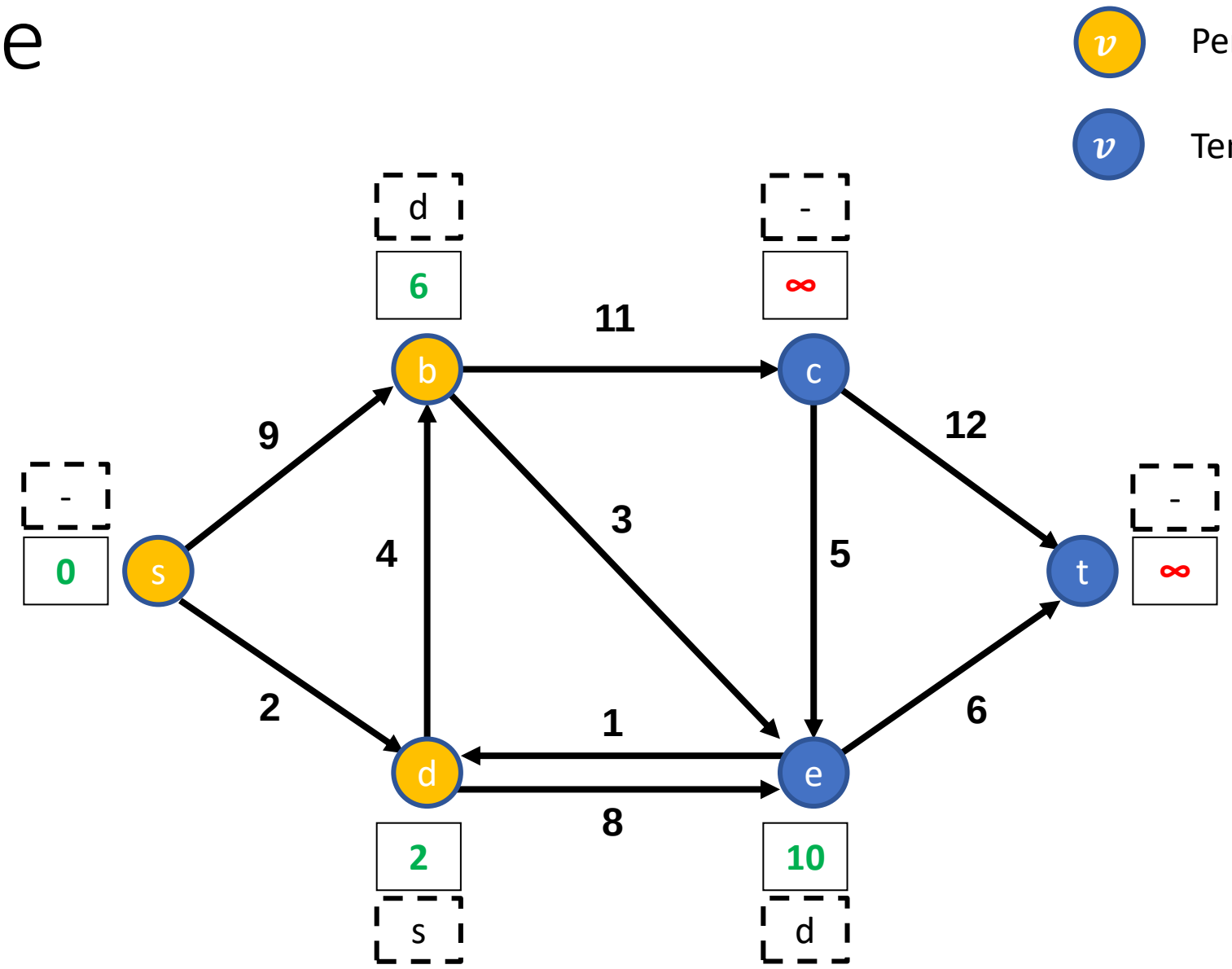



Update distances to reachable vertices which are labelled temporary

Sensitivity: Internal

# Example

| $v$ | Label | $d_v$ | $\pi_v$ |
|---|---|---|---|
| s | P | 0 | - |
| b | P | ~~9~~ 6 | ~~s~~ d |
| c | T | | |
| d | P | 2 | s |
| e | T | 10 | d |
| t | T | | |

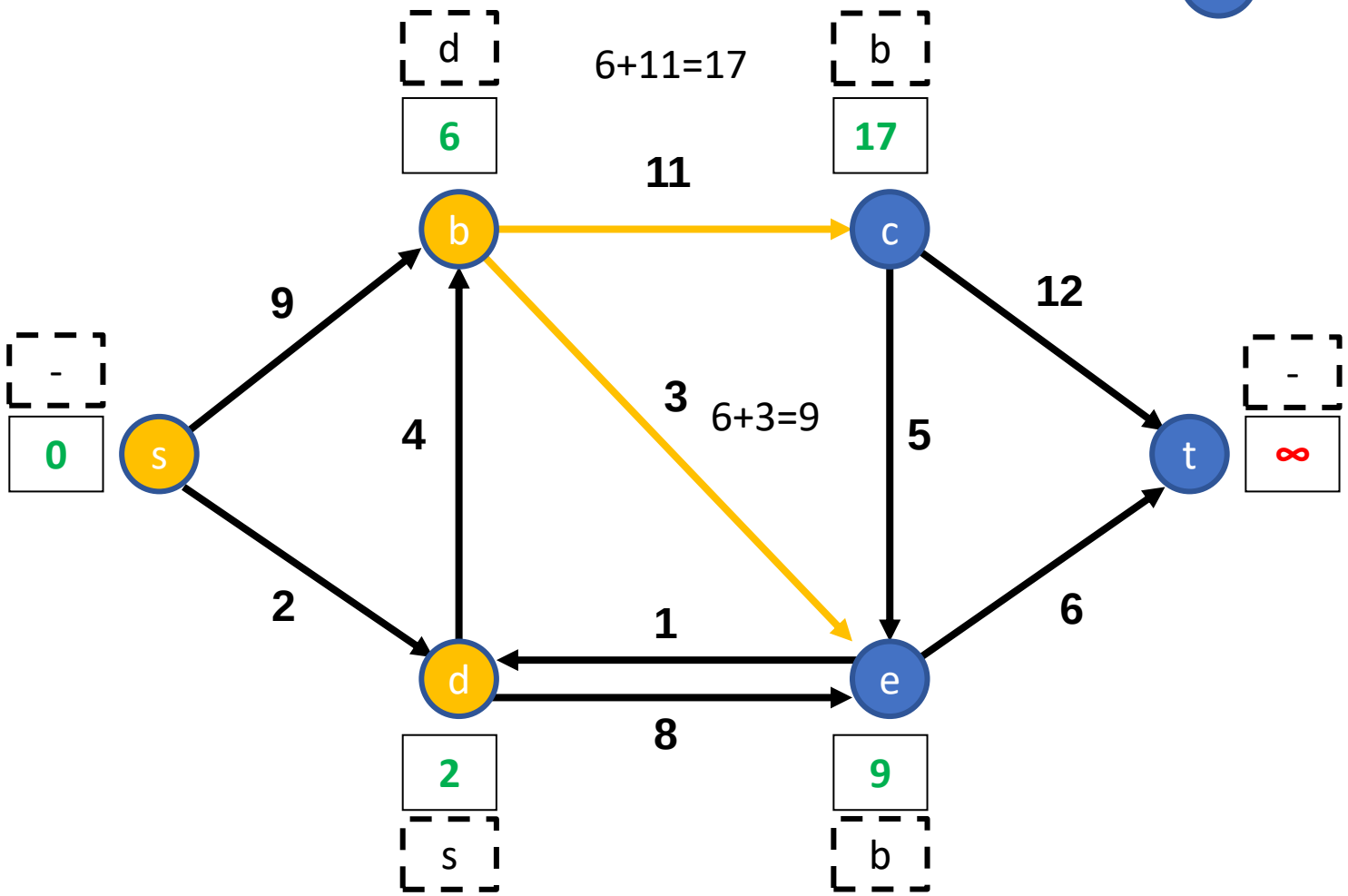



Label temporary vertex with minimum distance as permanent

Sensitivity: Internal

# Example

| $v$ | Label | $d_v$ | $\pi_v$ |
|---|---|---|---|
| s | P | 0 | - |
| b | P | ~~9~~ 6 | ~~s~~ d |
| c | T | 17 | b |
| d | P | 2 | s |
| e | T | ~~10~~ 9 | ~~d~~ b |
| t | T | | |

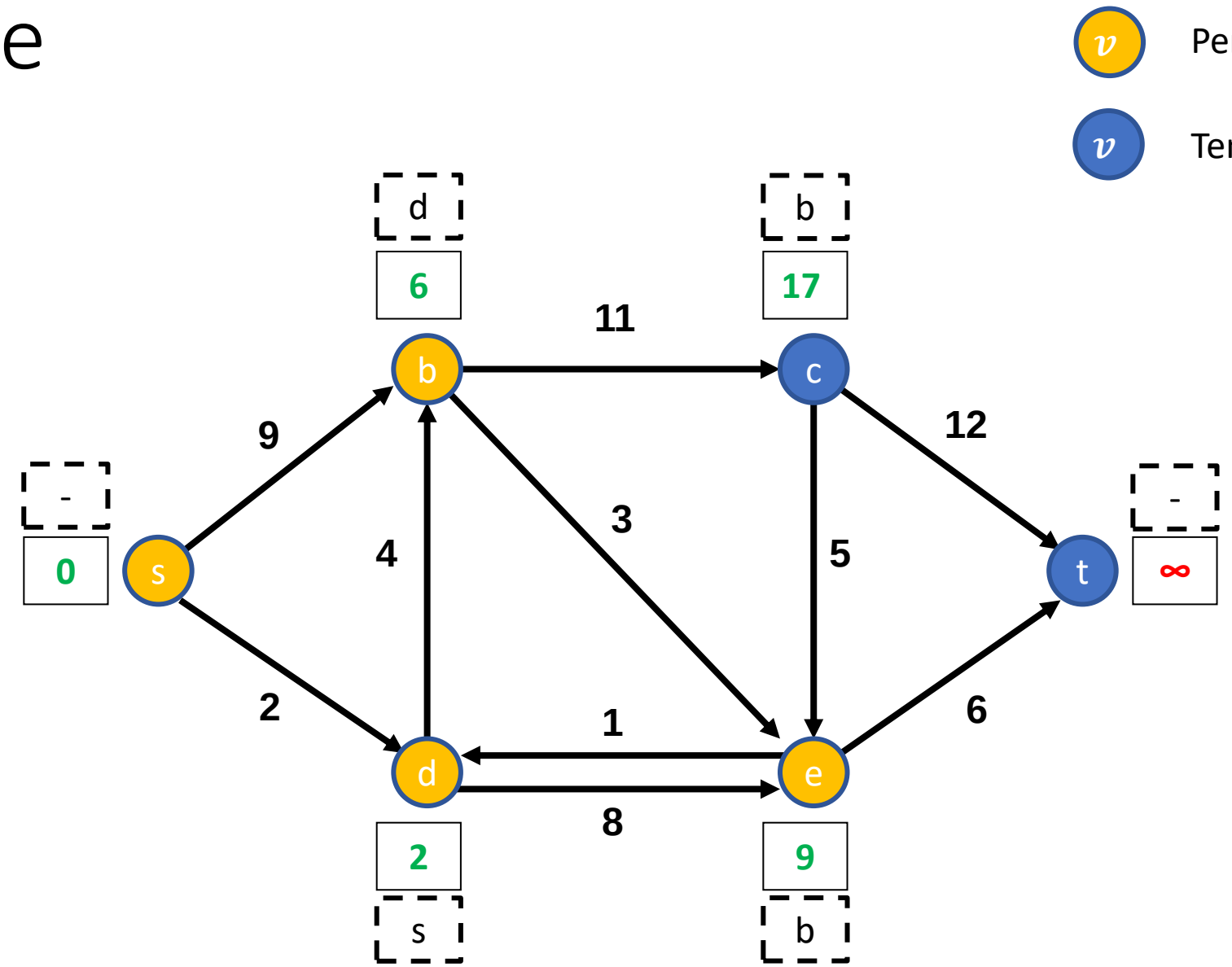



Update distances to reachable vertices which are labelled temporary

Sensitivity: Internal

# Example

| $v$ | Label | $d_v$ | $\pi_v$ |
|---|---|---|---|
| s | P | 0 | - |
| b | P | ~~9~~ 6 | ~~s~~ d |
| c | T | 17 | b |
| d | P | 2 | s |
| e | P | ~~10~~ 9 | ~~d~~ b |
| t | T | | |

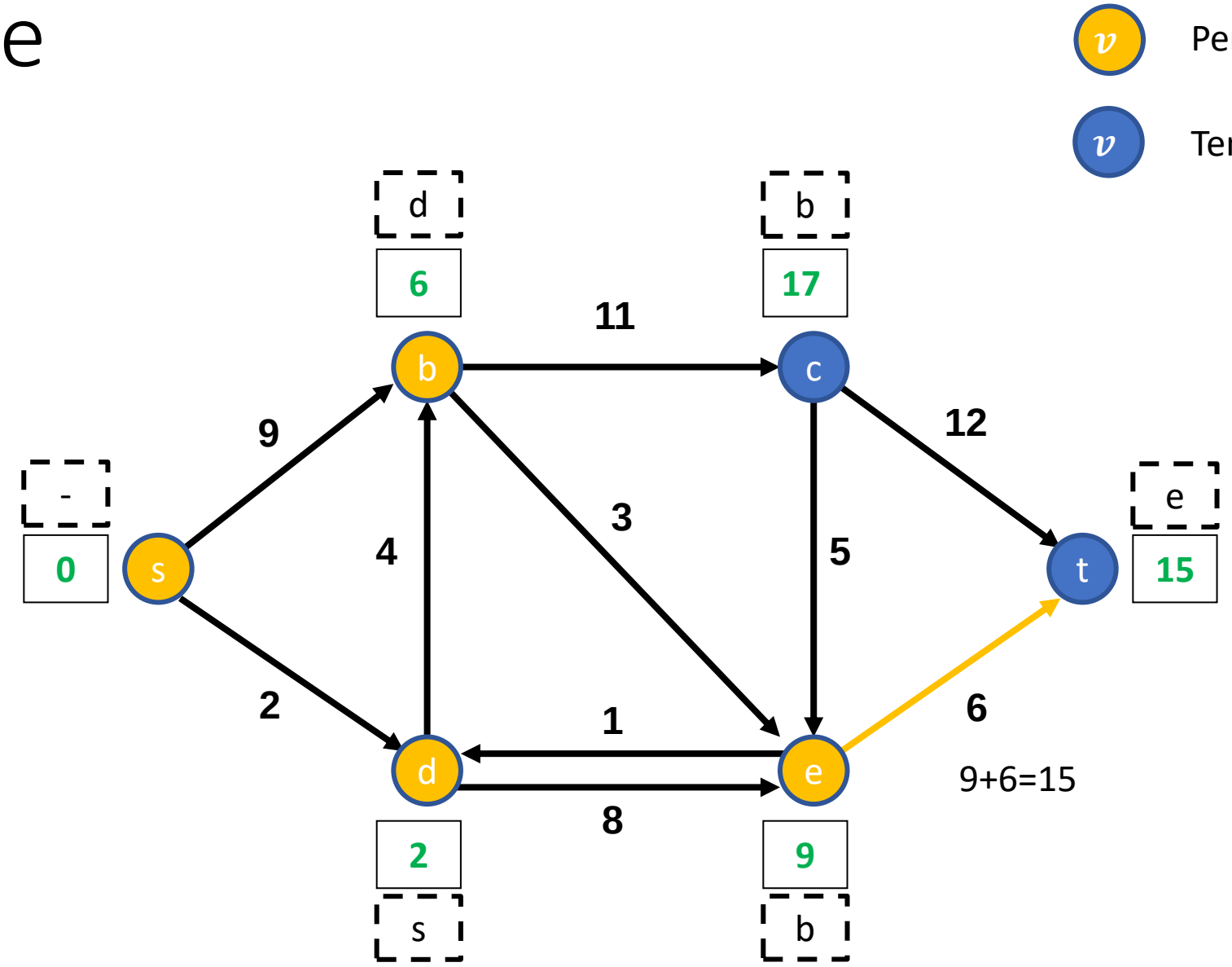



Label temporary vertex with minimum distance as permanent

Sensitivity: Internal

# Example

| $v$ | Label | $d_v$ | $\pi_v$ |
|---|---|---|---|
| s | P | 0 | - |
| b | P | ~~9~~ 6 | ~~s~~ d |
| c | T | 17 | b |
| d | P | 2 | s |
| e | P | ~~10~~ 9 | ~~d~~ b |
| t | T | 15 | e |

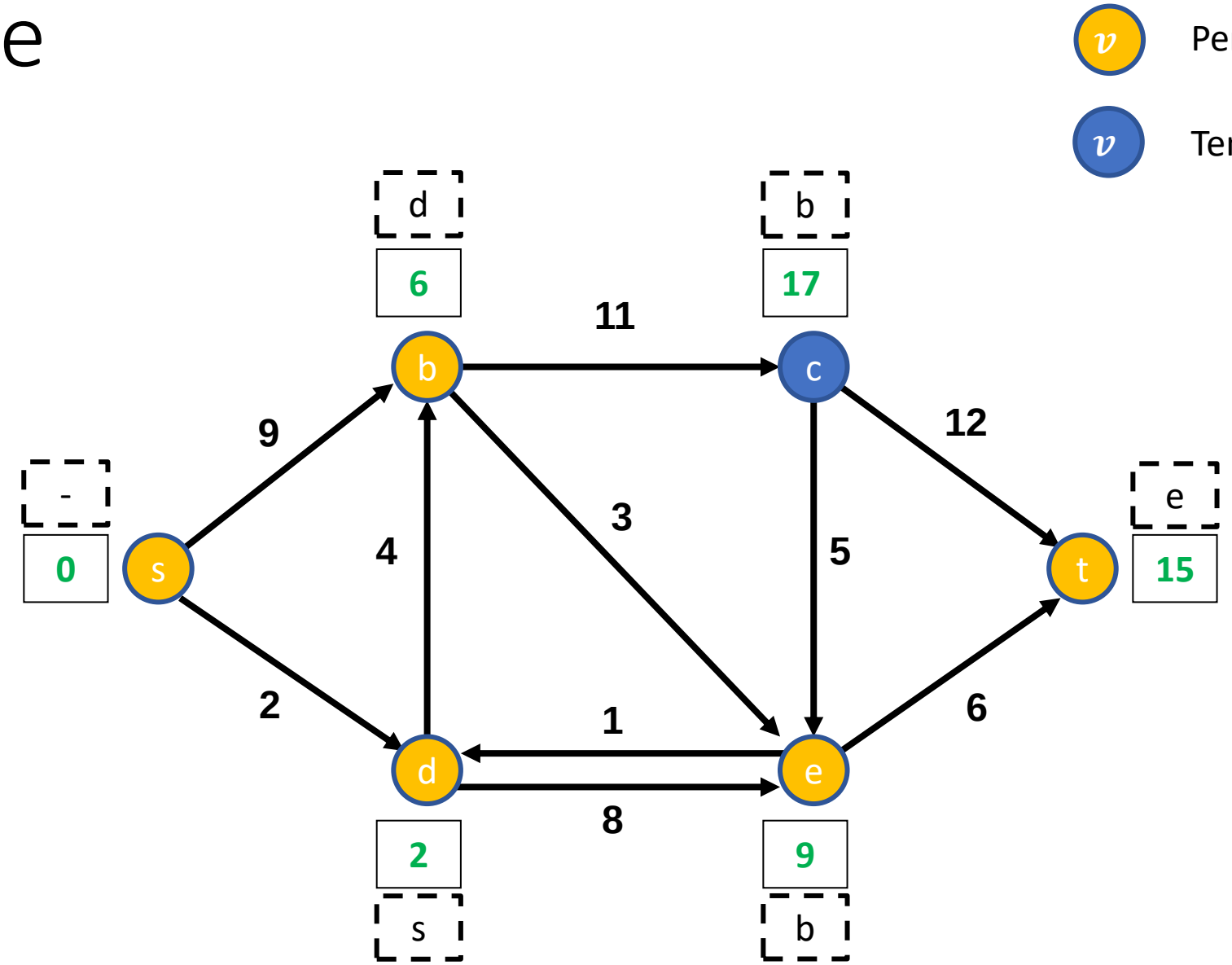



Update distances to reachable vertices which are labelled temporary

Sensitivity: Internal

# Example

| $v$ | Label | $d_v$ | $\pi_v$ |
|---|---|---|---|
| s | P | 0 | - |
| b | P | ~~9~~ 6 | ~~s~~ d |
| c | T | 17 | b |
| d | P | 2 | s |
| e | P | ~~10~~ 9 | ~~d~~ b |
| t | P | 15 | e |

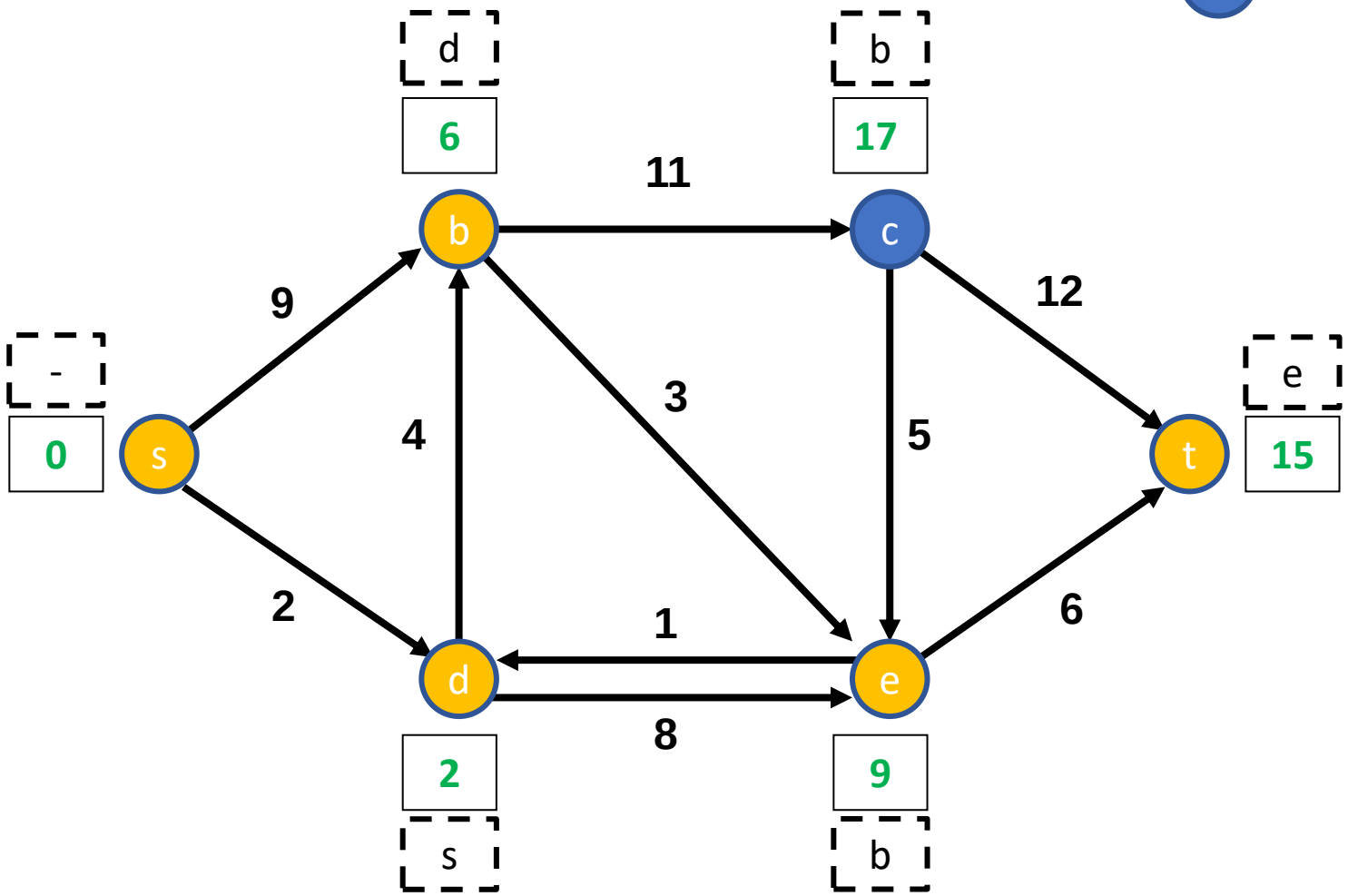



Label temporary vertex with minimum distance as permanent

Sensitivity: Internal

# Example

| $v$ | Label | $d_v$ | $\pi_v$ |
|---|---|---|---|
| s | P | 0 | - |
| b | P | ~~9~~ 6 | ~~s~~ d |
| c | T | 17 | b |
| d | P | 2 | s |
| e | P | ~~10~~ 9 | ~~d~~ b |
| t | P | 15 | e |

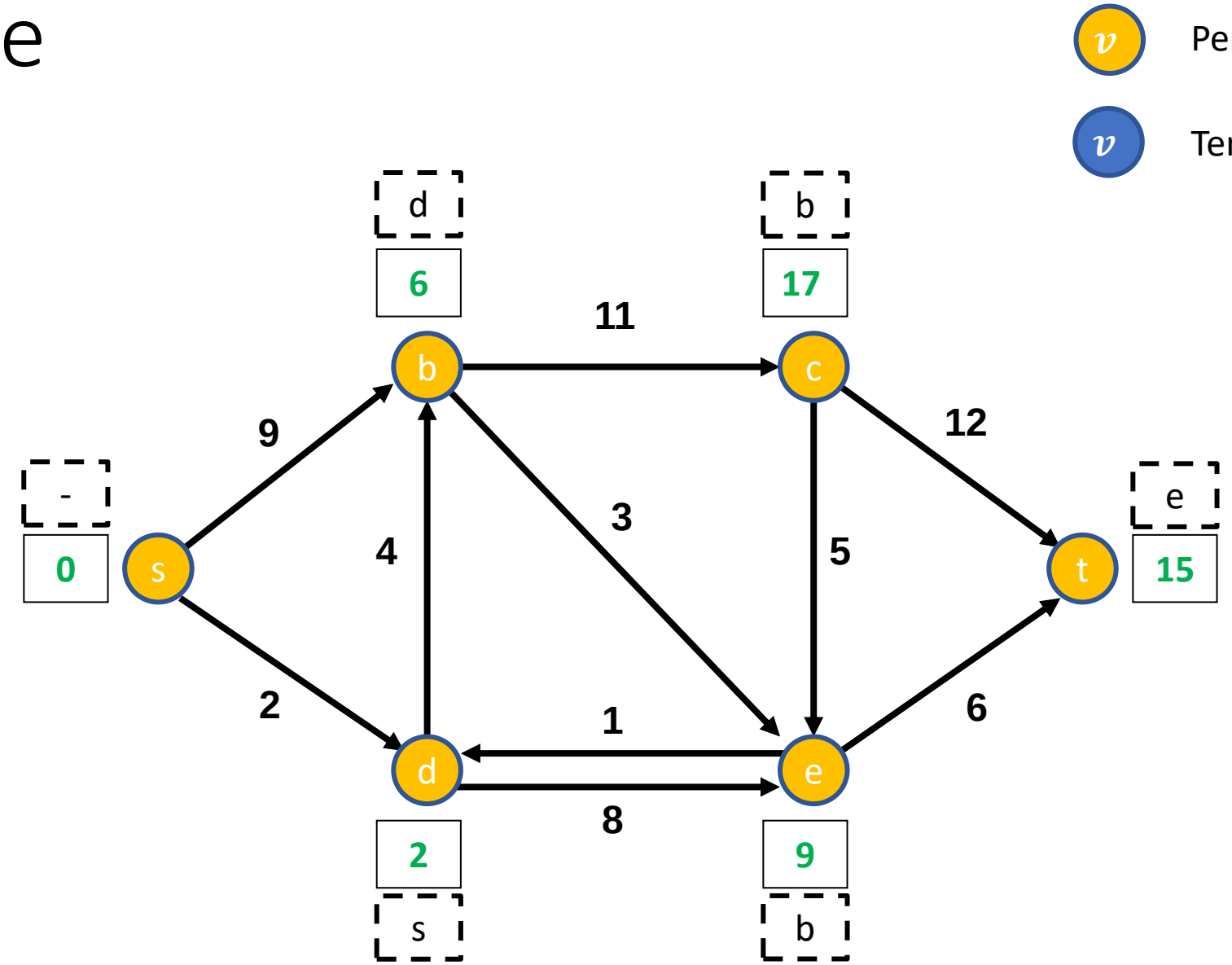



Update distances to reachable vertices which are labelled temporary. No changes

Sensitivity: Internal

# Example

| $v$ | Label | $d_v$ | $\pi_v$ |
|---|---|---|---|
| s | P | 0 | - |
| b | P | ~~9~~ 6 | ~~s~~ d |
| c | P | 17 | b |
| d | P | 2 | s |
| e | P | ~~10~~ 9 | ~~d~~ b |
| t | P | 15 | e |

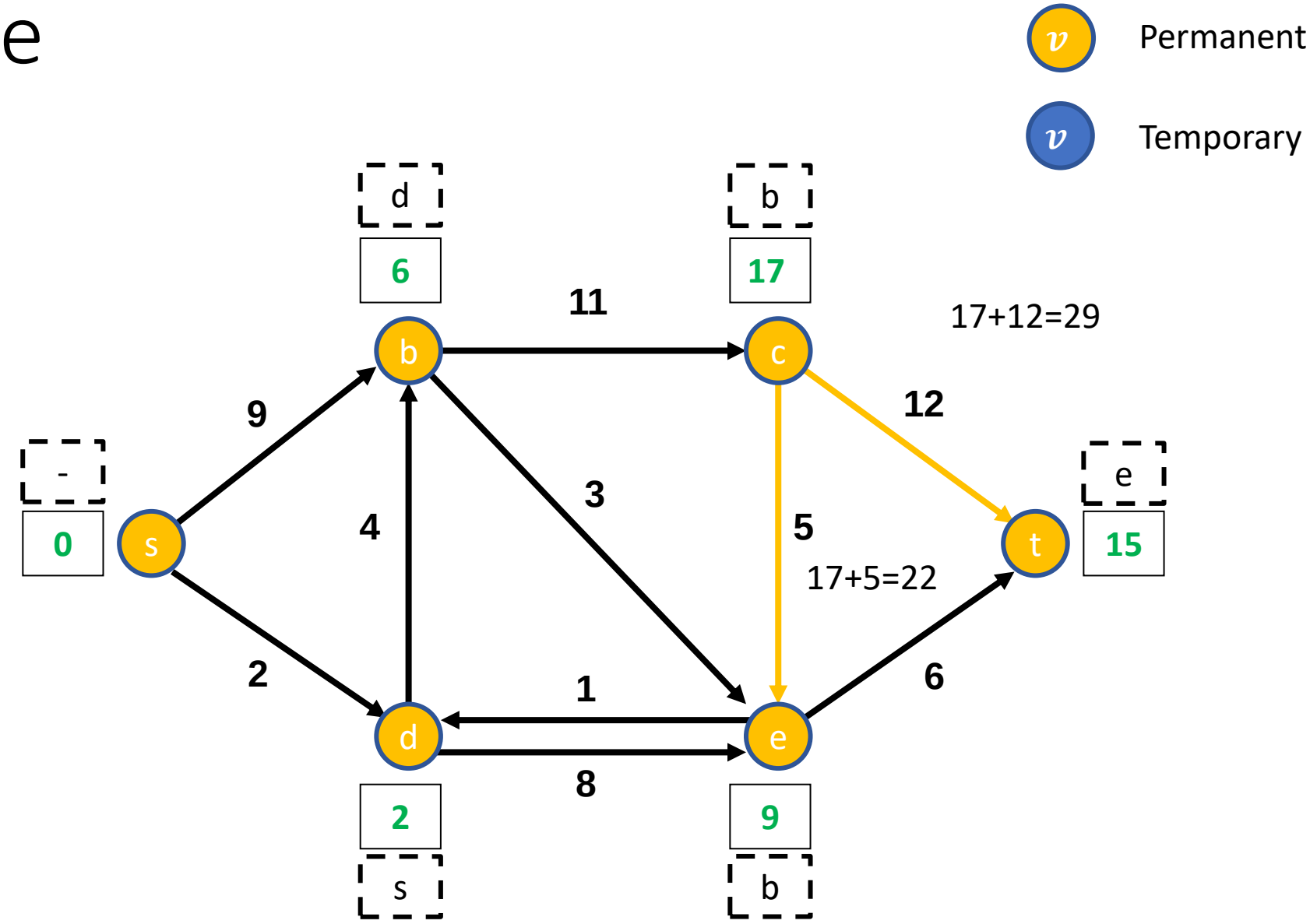



Label temporary vertex with minimum distance as permanent

Sensitivity: Internal

# Example

| $v$ | Label | $d_v$ | $\pi_v$ |
|---|---|---|---|
| s | P | 0 | - |
| b | P | ~~9~~ 6 | ~~s~~ d |
| c | P | 17 | b |
| d | P | 2 | s |
| e | P | ~~10~~ 9 | ~~d~~ b |
| t | P | 15 | e |

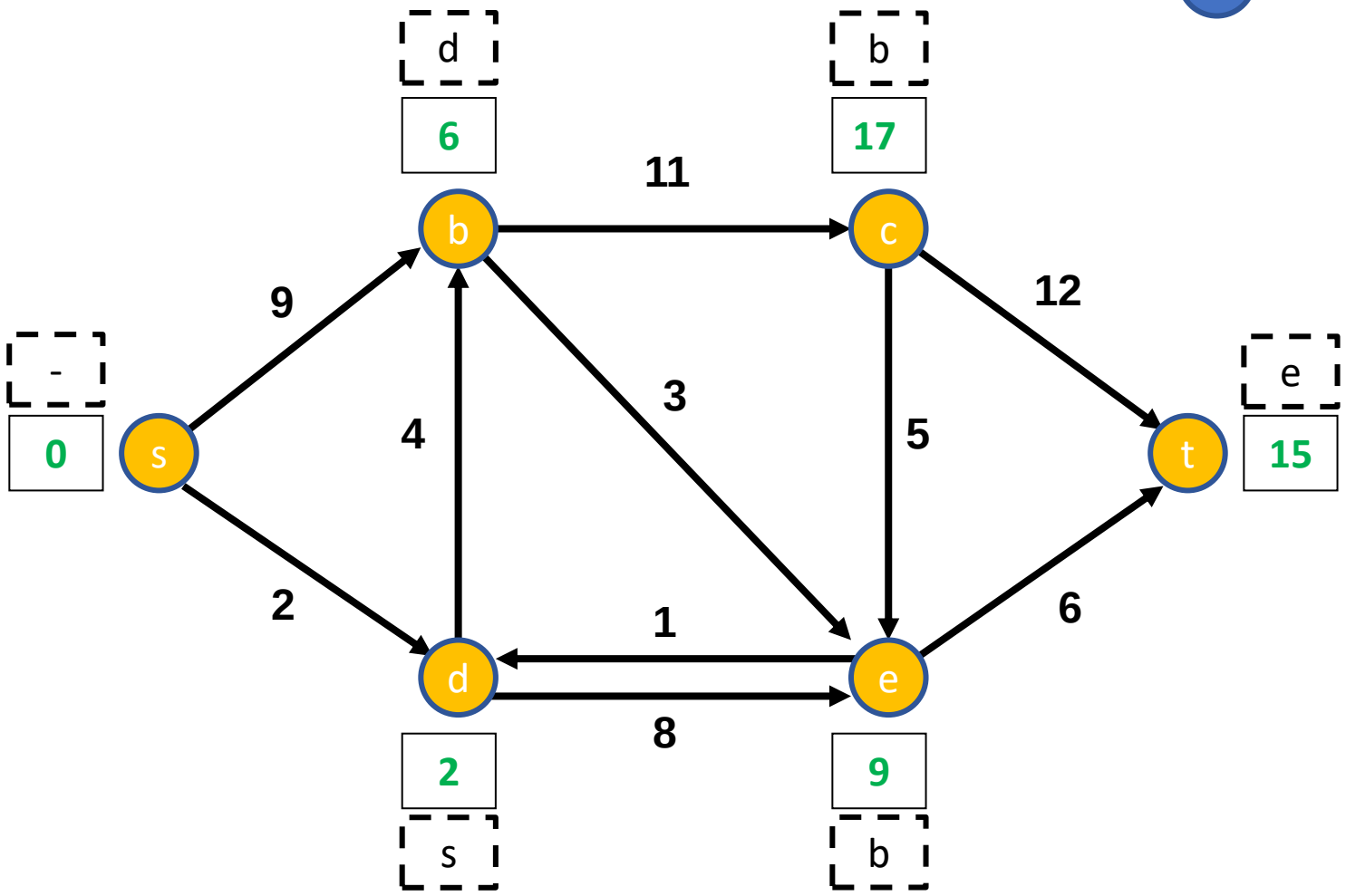



Update distances to reachable vertices which are labelled temporary. No changes

Sensitivity: Internal

# Example

| $v$ | Label | $d_v$ | $\pi_v$ |
|---|---|---|---|
| s | P | 0 | - |
| b | P | ~~9~~ 6 | ~~s~~ d |
| c | P | 17 | b |
| d | P | 2 | s |
| e | P | ~~10~~ 9 | ~~d~~ b |
| t | P | 15 | e |

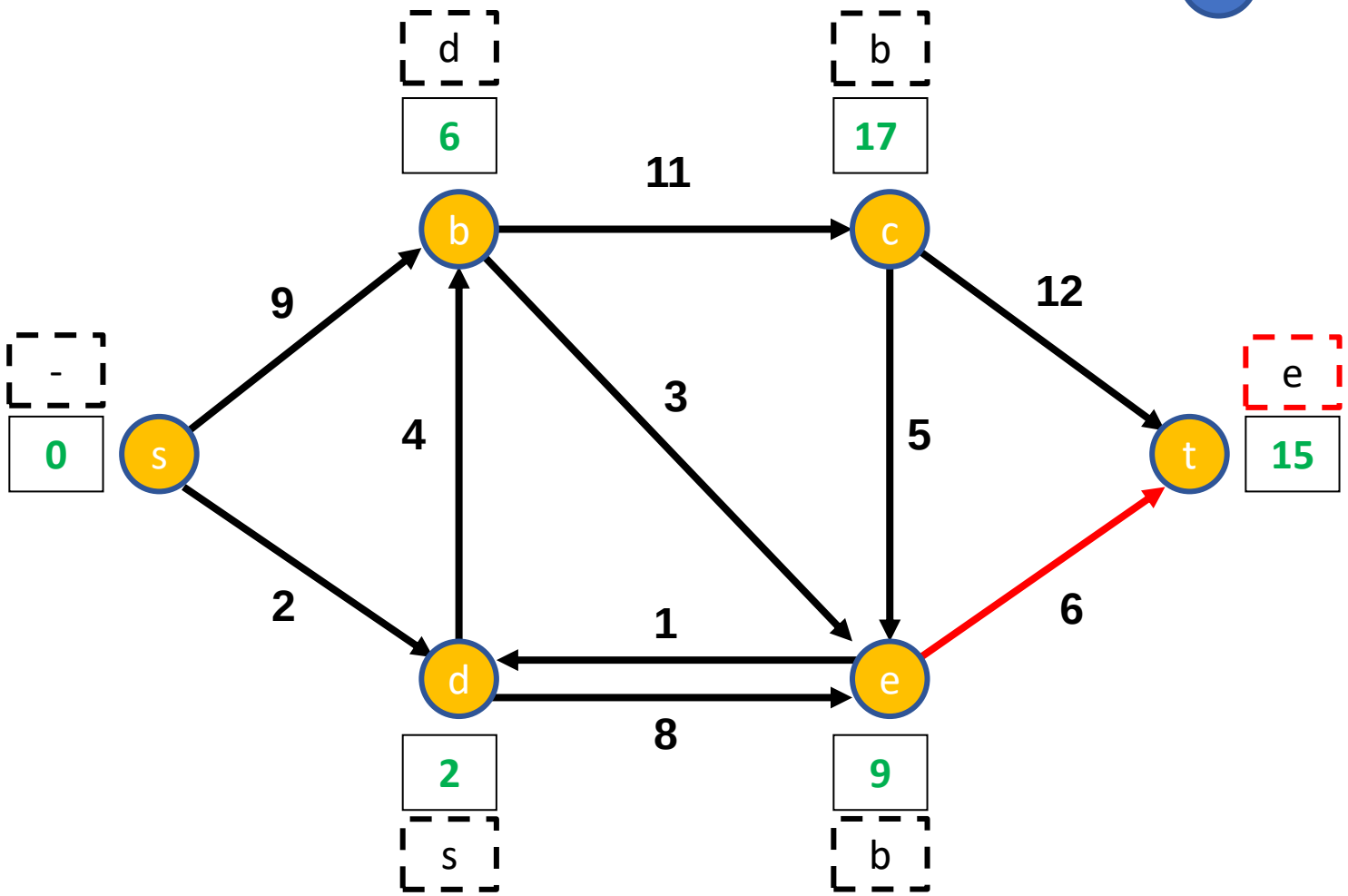



Final Network and Table

Sensitivity: Internal

# Example

| $v$ | Label | $d_v$ | $\pi_v$ |
|---|---|---|---|
| s | P | 0 | - |
| b | P | ~~9~~ 6 | ~~s~~ d |
| c | P | 17 | b |
| d | P | 2 | s |
| e | P | ~~10~~ 9 | ~~d~~ b |
| t | P | 15 | e |

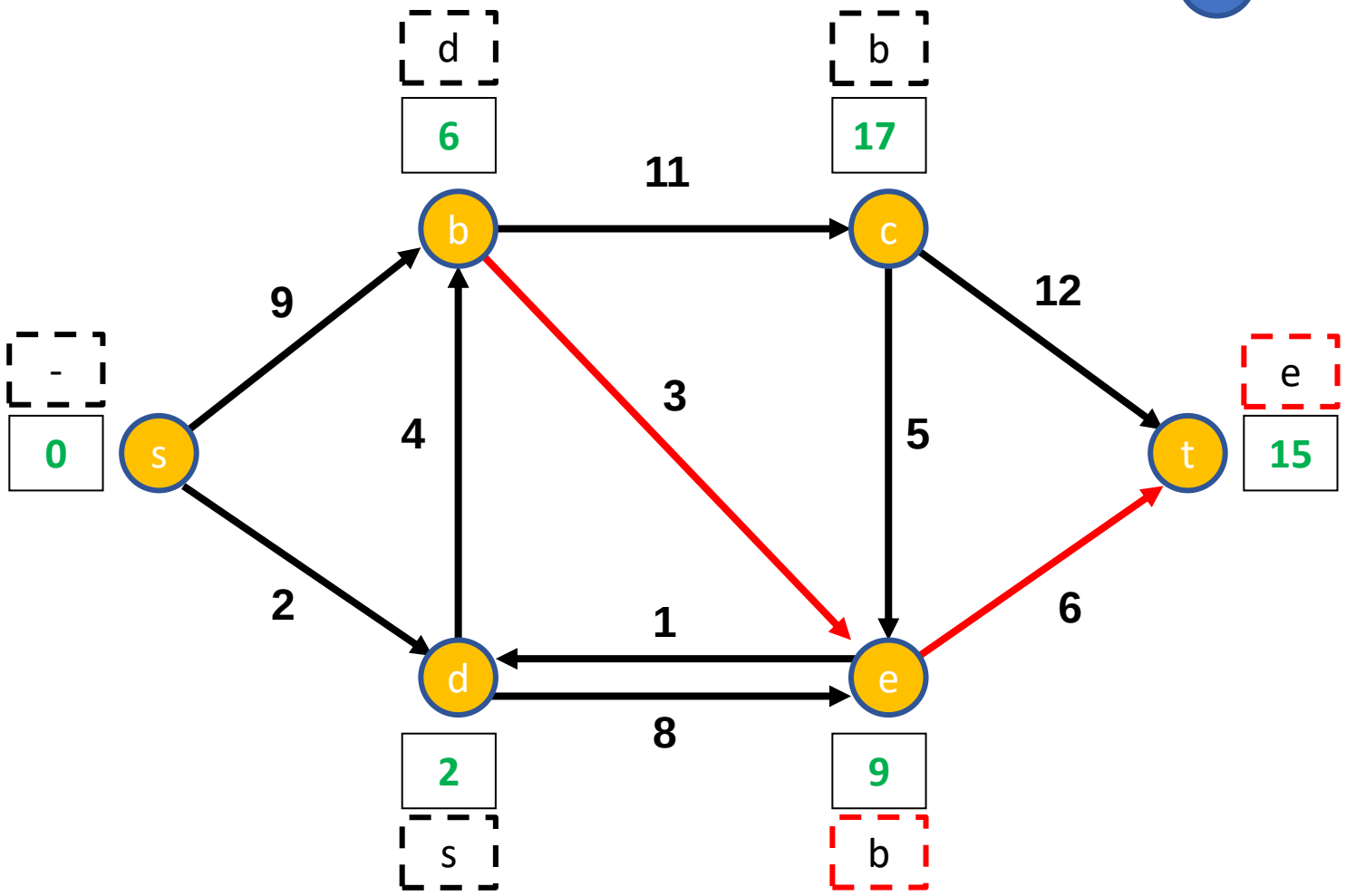



To compute the shortest path, work backwards from the destination t along the predecessors

Sensitivity: Internal

# Example

| $v$ | Label | $d_v$ | $\pi_v$ |
|---|---|---|---|
| s | P | 0 | - |
| b | P | ~~9~~ 6 | ~~s~~ d |
| c | P | 17 | b |
| d | P | 2 | s |
| e | P | ~~10~~ 9 | ~~d~~ b |
| t | P | 15 | e |

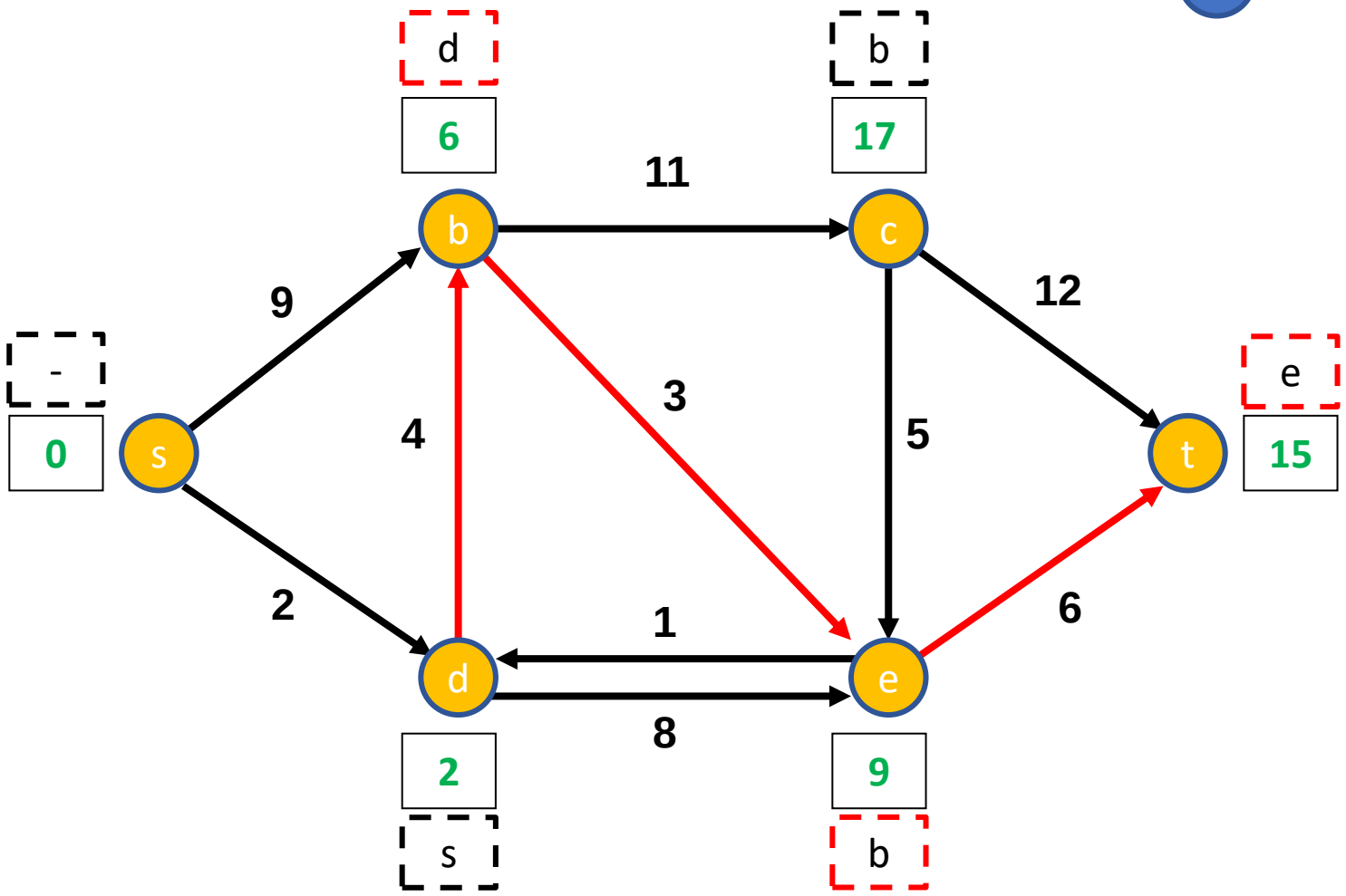



To compute the shortest path, work backwards from the destination t along the predecessors

Sensitivity: Internal

# Example

| $v$ | Label | $d_v$ | $\pi_v$ |
|---|---|---|---|
| s | P | 0 | - |
| b | P | ~~9~~ 6 | ~~s~~ d |
| c | P | 17 | b |
| d | P | 2 | s |
| e | P | ~~10~~ 9 | ~~d~~ b |
| t | P | 15 | e |

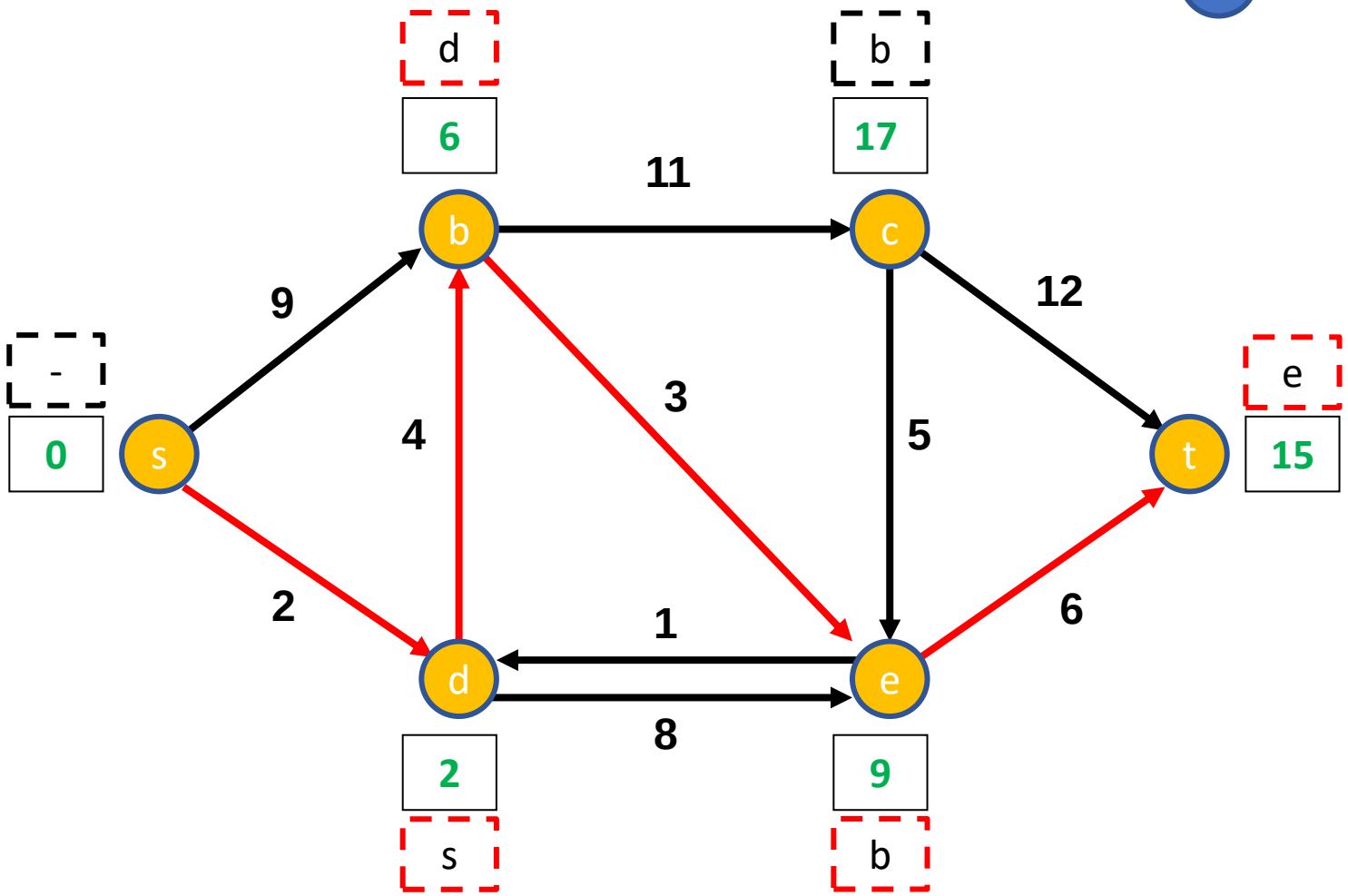



To compute the shortest path, work backwards from the destination t along the predecessors

Sensitivity: Internal

# Example

| $v$ | Label | $d_v$ | $\pi_v$ |
|---|---|---|---|
| s | P | 0 | - |
| b | P | ~~9~~ 6 | ~~s~~ d |
| c | P | 17 | b |
| d | P | 2 | s |
| e | P | ~~10~~ 9 | ~~d~~ b |
| t | P | 15 | e |

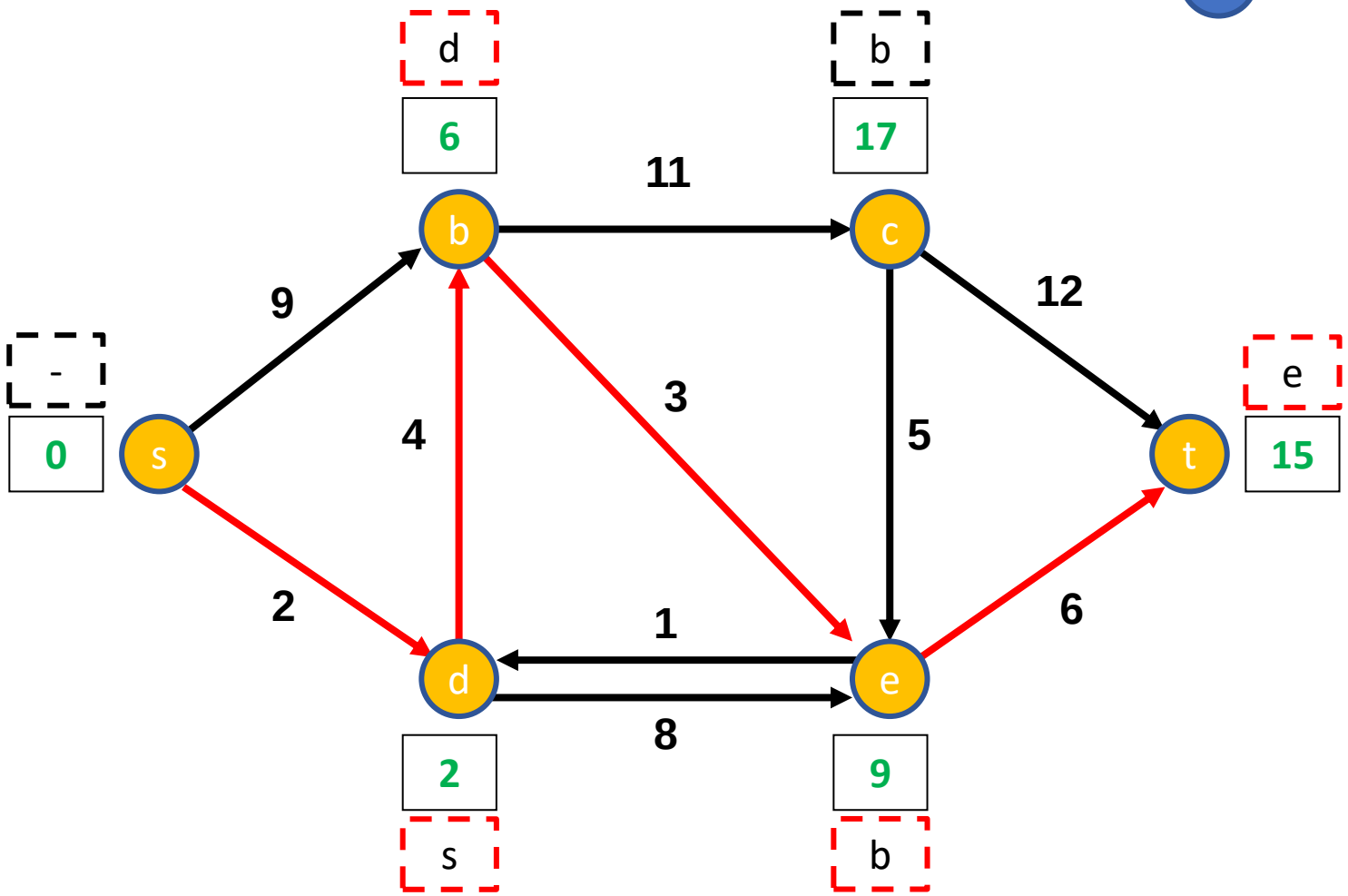



To compute the shortest path, work backwards from the destination t along the predecessors

Sensitivity: Internal

# Example

| $v$ | Label | $d_v$ | $\pi_v$ |
|---|---|---|---|
| s | P | 0 | - |
| b | P | ~~9~~ 6 | ~~s~~ d |
| c | P | 17 | b |
| d | P | 2 | s |
| e | P | ~~10~~ 9 | ~~d~~ b |
| t | P | 15 | e |




Shortest path from $s$ to $t$ is $s - d - b - e - t$ with a path length of 15

Sensitivity: Internal

# Correctness

Visits all vertices in increasing order of distance and visits each vertex exactly once.

Dijkstra maintains a set of visited vertices (permanent) and unvisited vertices (temporary).

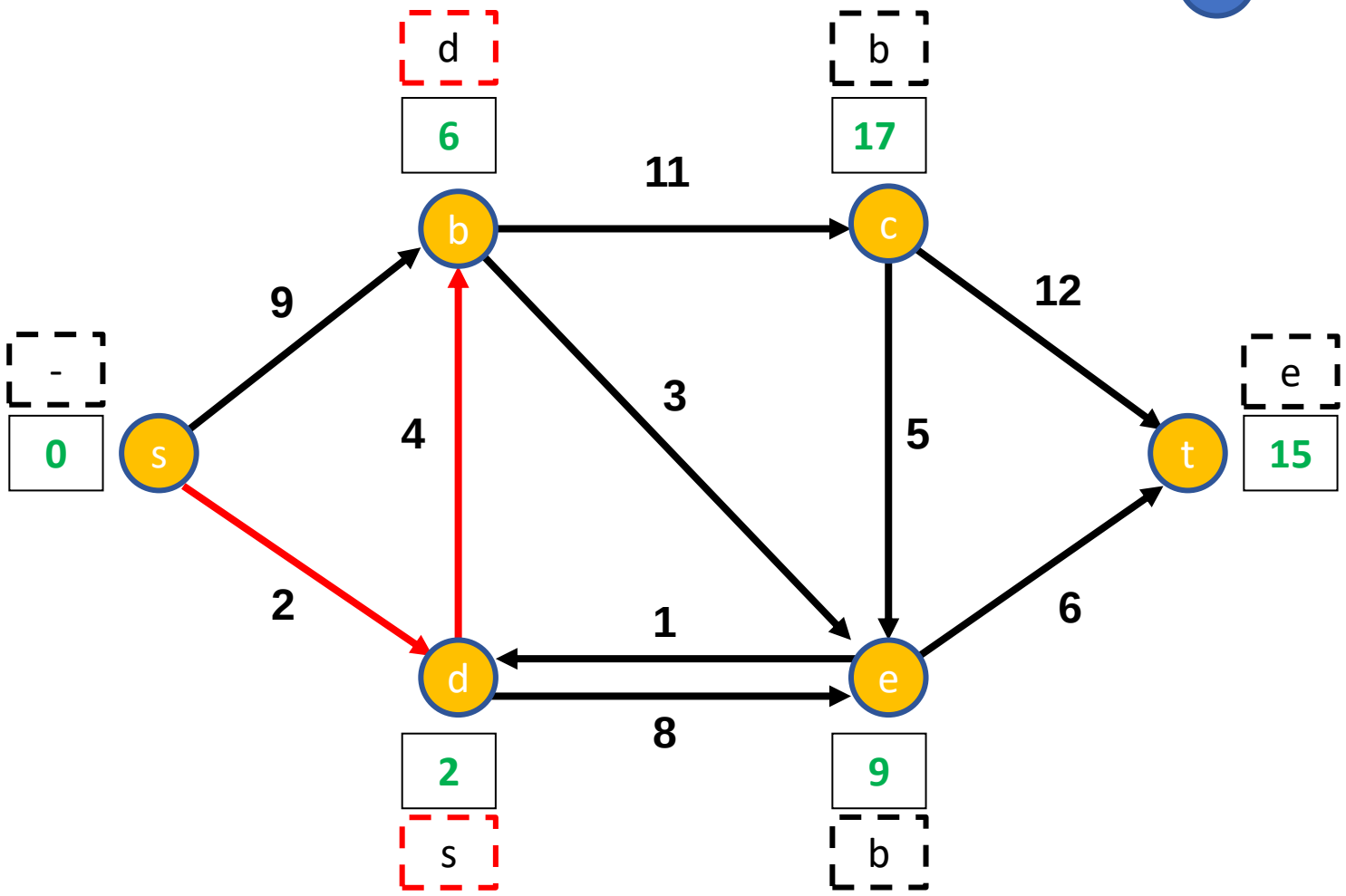
**Claim**: For a visited vertex $v$, $d_v$ and $\pi_v$ will be the optimal shortest path distance and corresponding predecessor vertex.

Meaning, we get all the shortest paths from the source to all visited vertices free!

Sensitivity: Internal

# Shortest sub-path

| $v$ | Label | $d_v$ | $\pi_v$ |
|---|---|---|---|
| s | P | 0 | - |
| b | P | ~~9~~ 6 | ~~s~~ d |
| c | P | 17 | b |
| d | P | 2 | s |
| e | P | ~~10~~ 9 | ~~d~~ b |
| t | P | 15 | e |




For example the shortest path from vertex $s$ to $b$ is $s - d - b$ of length 6

Sensitivity: Internal

# Proof and Time Complexity

Dijkstra's Algorithm has a worst-case time complexity of $O(V^2)$

- $O(|V|^2)$ with a simple array (scan every vertex for the minimum)
- $O((|E| + |V|) \log |V|)$ with an adjacency list and a binary heap

If you are interested you can find all the proofs about the correctness and time complexities of Dijkstra's and the other algorithms in:

Cormen, T. H., Leiserson, C. E., Rivest, R. L., Stein, C. (2001). *Introduction to Algorithms*. The MIT Press. ISBN: 0262032937

Or check out MIT 6.006 Lecture 16 (Dijkstra) in this week's 6 Further Reading folder.

Note they are somewhat complicated if you haven't seen this type of argument before.

Sensitivity: Internal

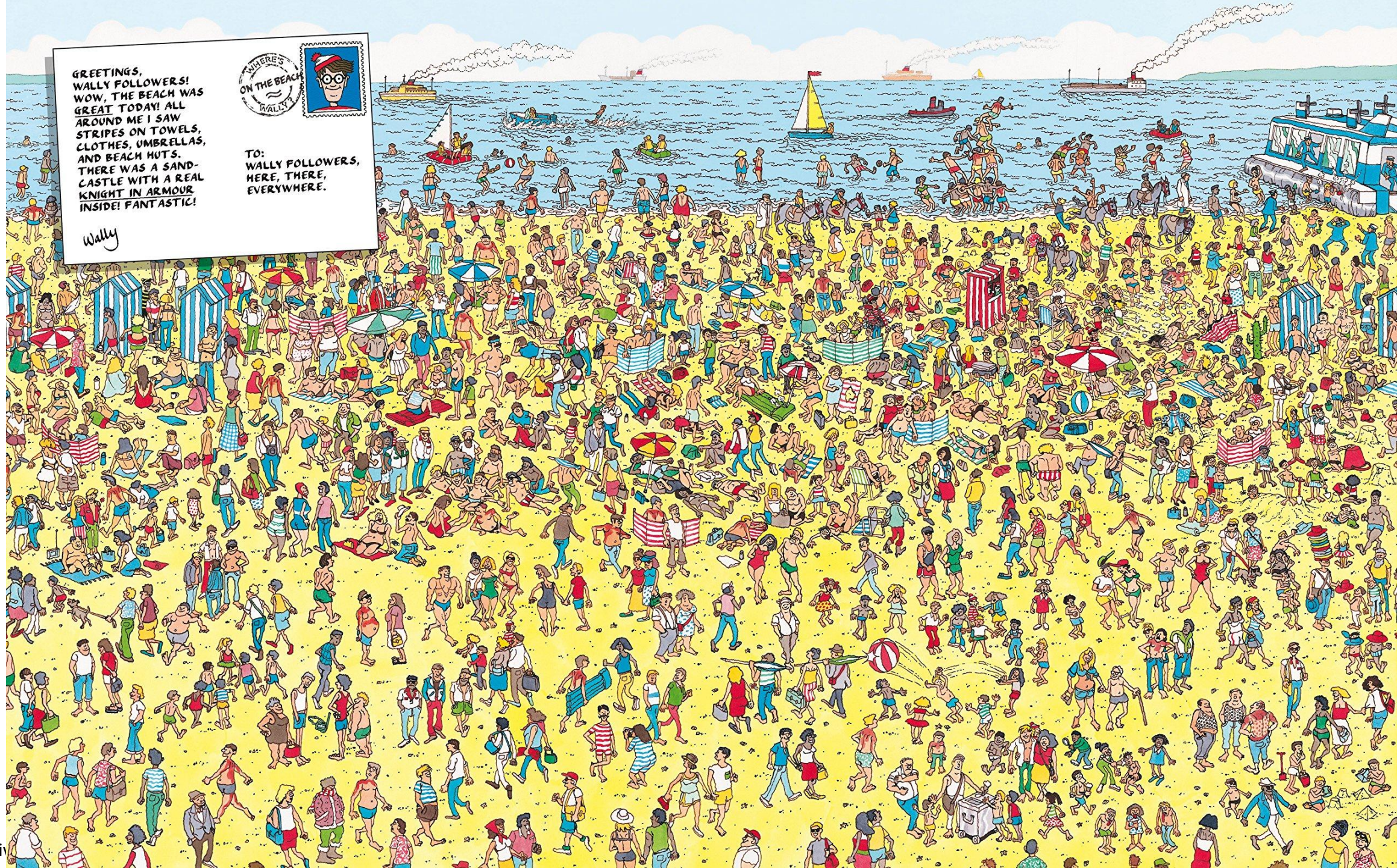
# 30 seconds....

You have 30 seconds to find Wally.

Raise your hand if you find him and I will pause to check.



Sensitivity: Internal

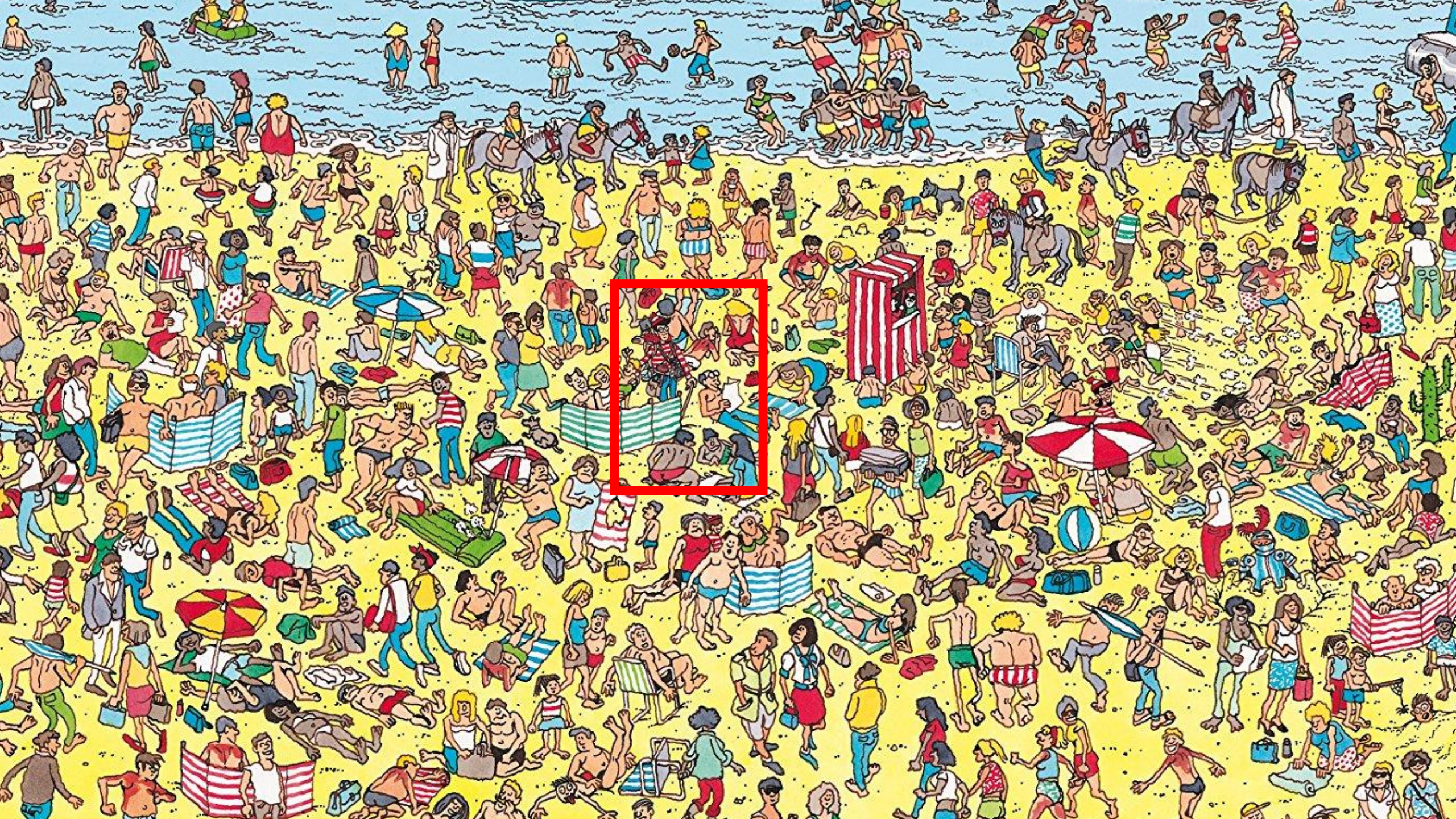

GREETINGS,
WALLY FOLLOWERS!
WOW, THE BEACH WAS
GREAT TODAY! ALL
AROUND ME I SAW
STRIPES ON TOWELS,
CLOTHES, UMBRELLAS,
AND BEACH HUTS.
THERE WAS A SAND-
CASTLE WITH A REAL
KNIGHT IN ARMOUR
INSIDE! FANTASTIC!
Wally
WHERE'S WALLY?
ON THE BEACH
TO:
WALLY FOLLOWERS,
HERE, THERE,
EVERYWHERE.

Sensitiv

# Checking Answer….

Please be patient, there is some high-tech stuff going on right now…

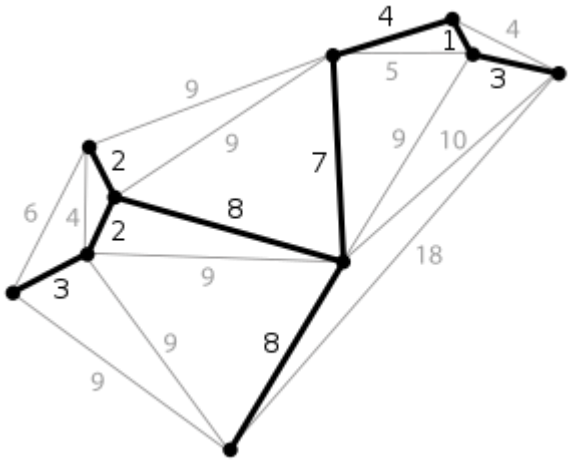
Sensitivity: Internal



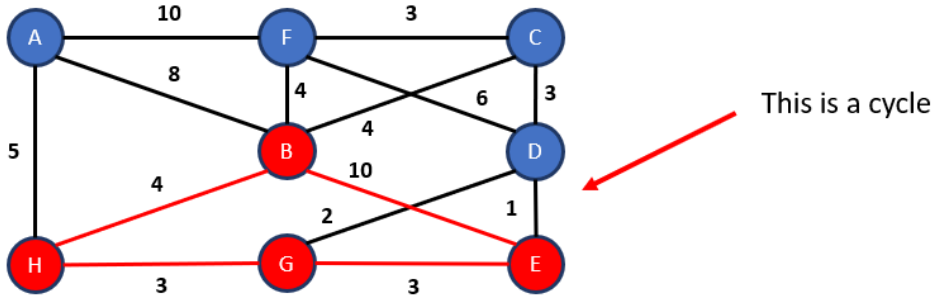
# Minimum Spanning Tree

**Definition**

A subset of weighted edges that connects all the vertices of a connected, weighted, undirected graph together without any cycles and has the minimum total edge weight.




Minimum Spanning Tree




Sensitivity: Internal

# Example Application

A fibre optics company is looking to expand its portfolio by providing broadband to an entirely new set of areas. Currently it has cable laid to a particular location.

The cable is extremely expensive and particularly difficult to lay in certain areas, therefore the company wishes to minimise the cost of laying the cable, whilst connecting all areas.

The company can lay cable between certain areas and has approximated the cost. Compute the minimum cost of connecting all the areas.

Sensitivity: Internal

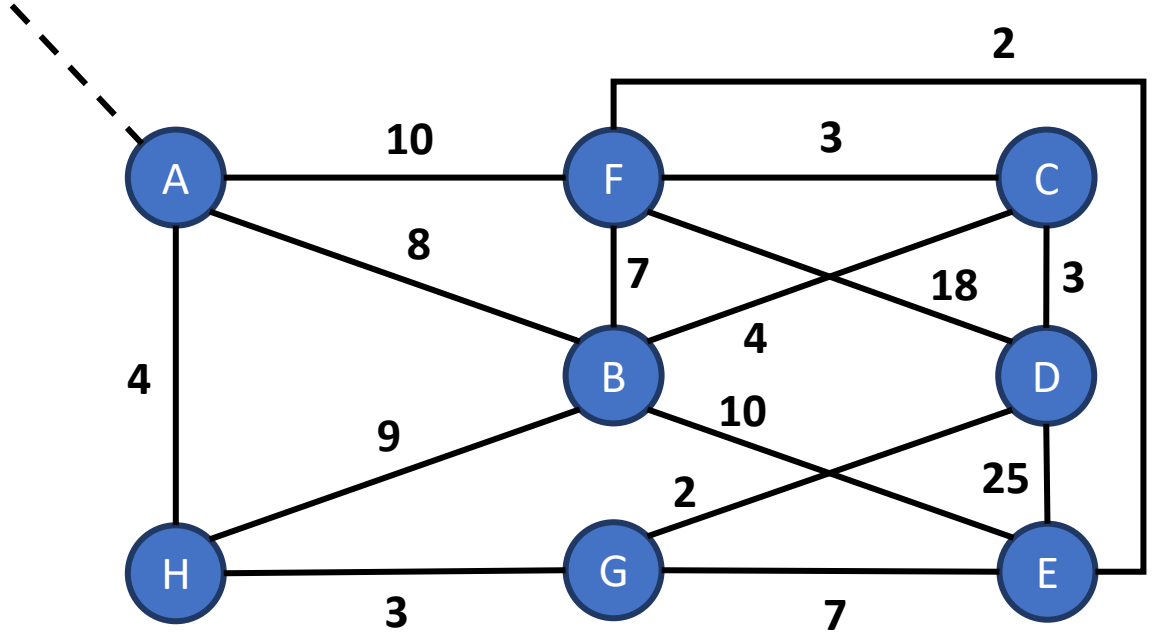
# Prim’s Algorithm

1. Initialise distances to vertices as $\infty$ and predecessors as $-$
2. Arbitrarily select a vertex and label as selected
3. Update distances and predecessors to adjacent unselected vertices
4. Select the reachable vertex with minimum distance
5. Repeat 3 and 4 until all vertices have been selected

Sensitivity: Internal

# Prim's Algorithm

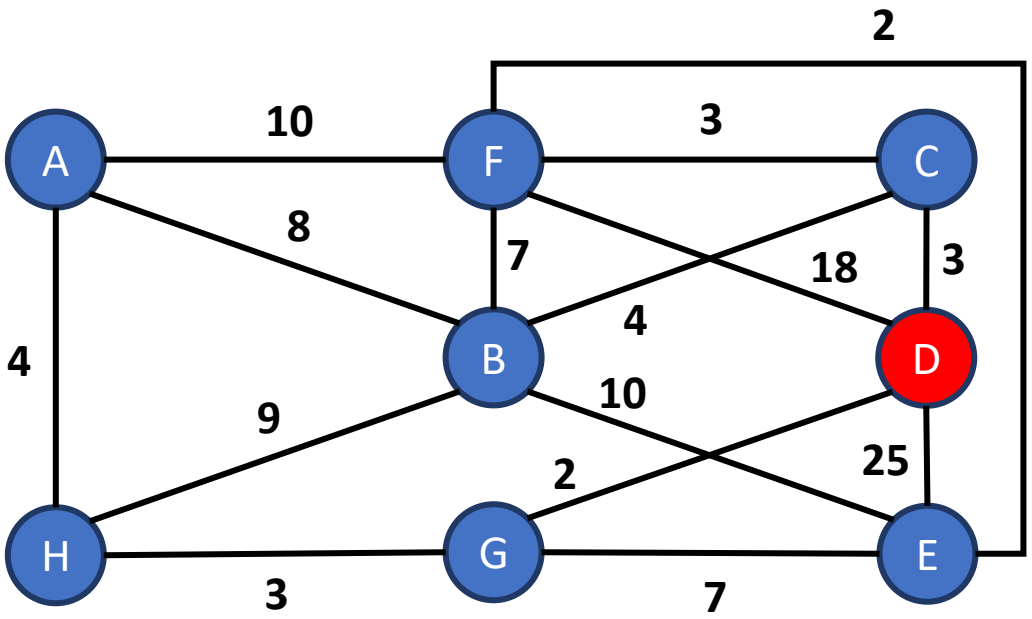
Existing Cable




| $v$ | Selected | $d_v$ | $\pi_v$ |
|---|---|---|---|
| A | | | |
| B | | | |
| C | | | |
| D | | | |
| E | | | |
| F | | | |
| G | | | |
| H | | | |

Sensitivity: Internal

# Prim's Algorithm

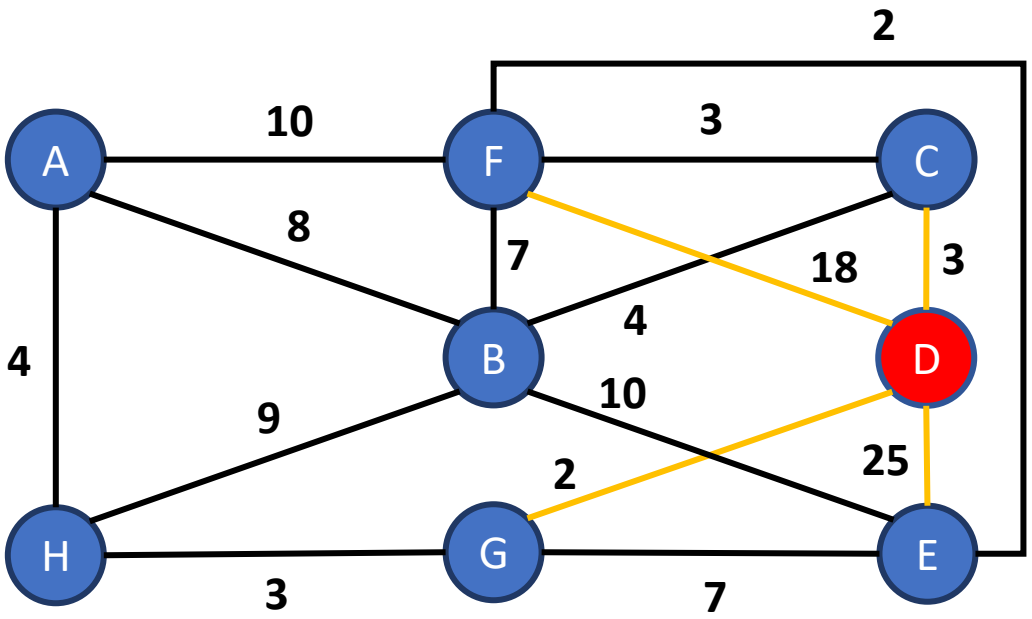



| $v$ | Selected | $d_v$ | $\pi_v$ |
|---|---|---|---|
| A | | | |
| B | | | |
| C | | | |
| D | T | 0 | - |
| E | | | |
| F | | | |
| G | | | |
| H | | | |

Arbitrarily select a vertex and mark as selected (here D; any start vertex gives the same total weight).

Sensitivity: Internal

# Prim's Algorithm

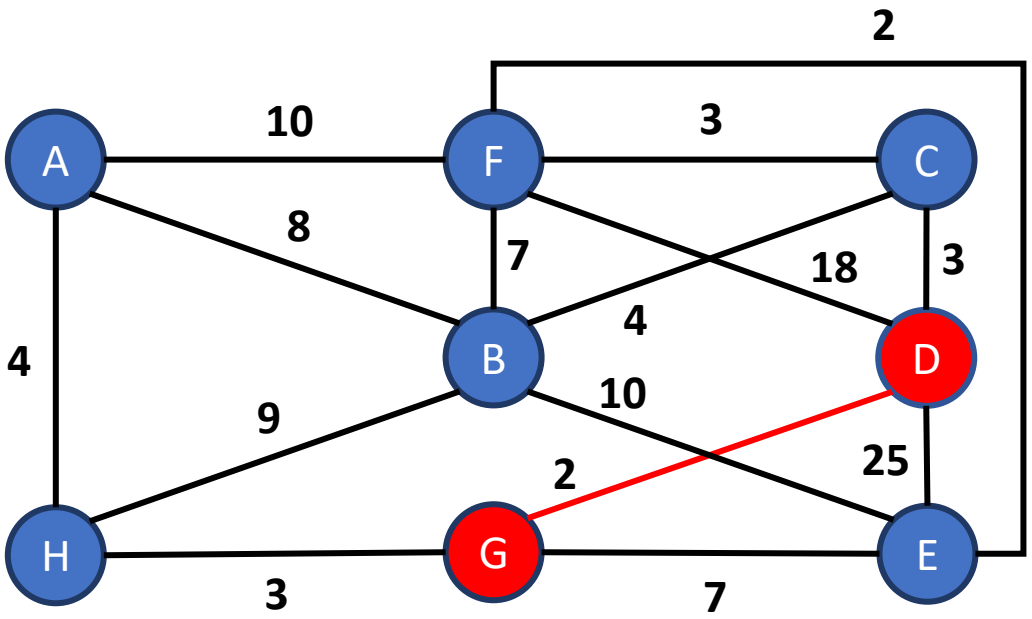



| $v$ | Selected | $d_v$ | $\pi_v$ |
|---|---|---|---|
| A | | | |
| B | | | |
| C | | 3 | D |
| D | T | 0 | - |
| E | | 25 | D |
| F | | 18 | D |
| G | | 2 | D |
| H | | | |

Update distances of adjacent, unselected vertices

Sensitivity: Internal

# Prim's Algorithm

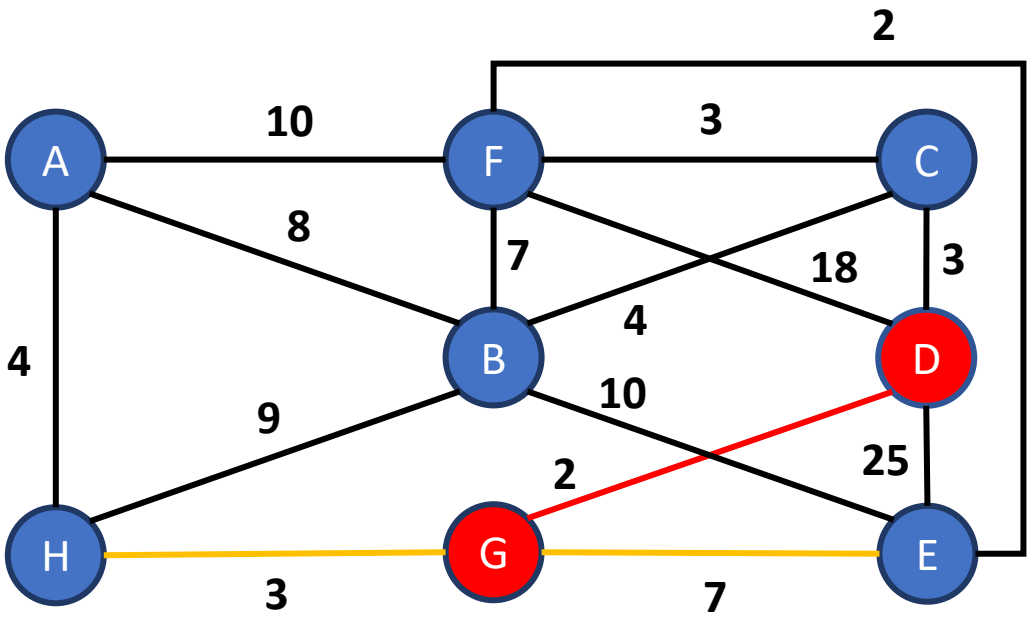



| $v$ | Selected | $d_v$ | $\pi_v$ |
|---|---|---|---|
| A | | | |
| B | | | |
| C | | 3 | D |
| D | T | 0 | - |
| E | | 25 | D |
| F | | 18 | D |
| G | T | 2 | D |
| H | | | |

Select reachable vertex with minimum distance

Sensitivity: Internal

# Prim's Algorithm

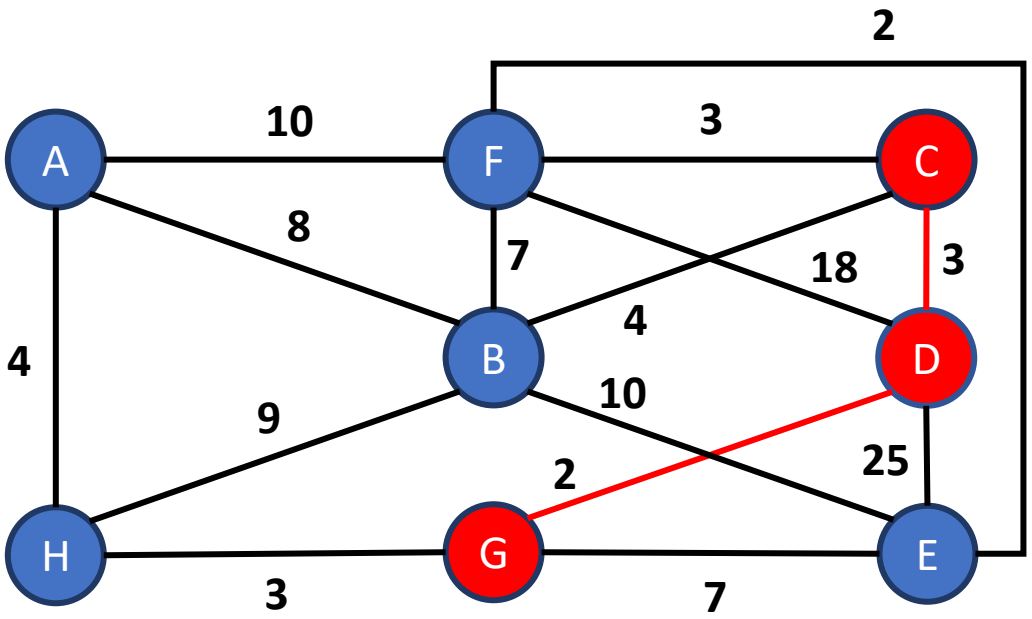



| $v$ | Selected | $d_v$ | $\pi_v$ |
|---|---|---|---|
| A | | | |
| B | | | |
| C | | 3 | D |
| D | T | 0 | - |
| E | | ~~25~~ 7 | ~~D~~ G |
| F | | 18 | D |
| G | T | 2 | D |
| H | | 3 | G |

Update distances of adjacent, unselected vertices

Sensitivity: Internal

# Prim's Algorithm

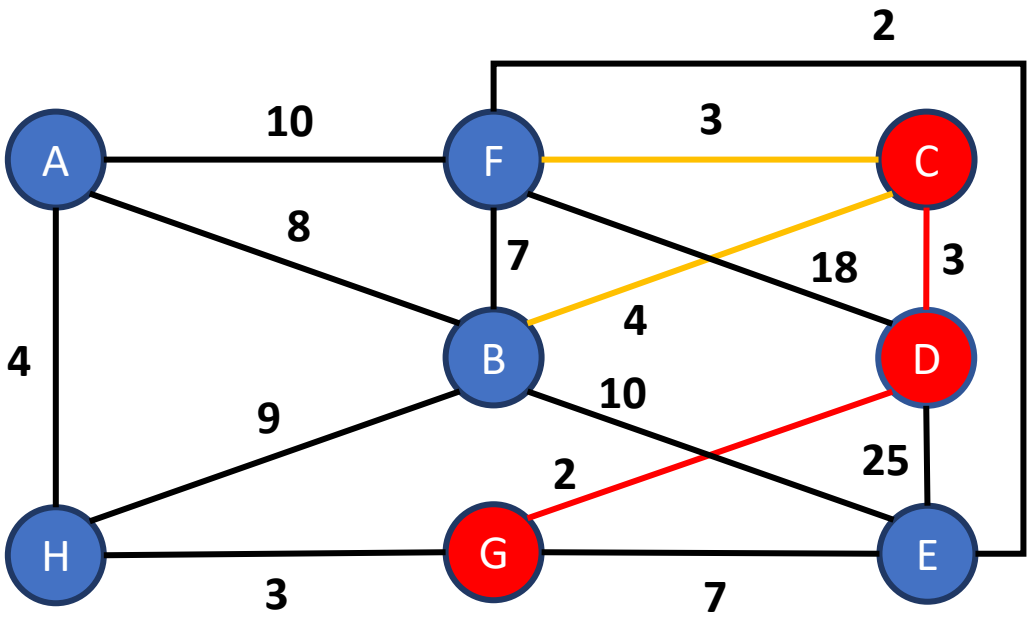



| $v$ | Selected | $d_v$ | $\pi_v$ |
|---|---|---|---|
| A | | | |
| B | | | |
| C | T | 3 | D |
| D | T | 0 | - |
| E | | ~~25~~ 7 | ~~D~~ G |
| F | | 18 | D |
| G | T | 2 | D |
| H | | 3 | G |

Select reachable vertex with minimum distance

Sensitivity: Internal

# Prim's Algorithm

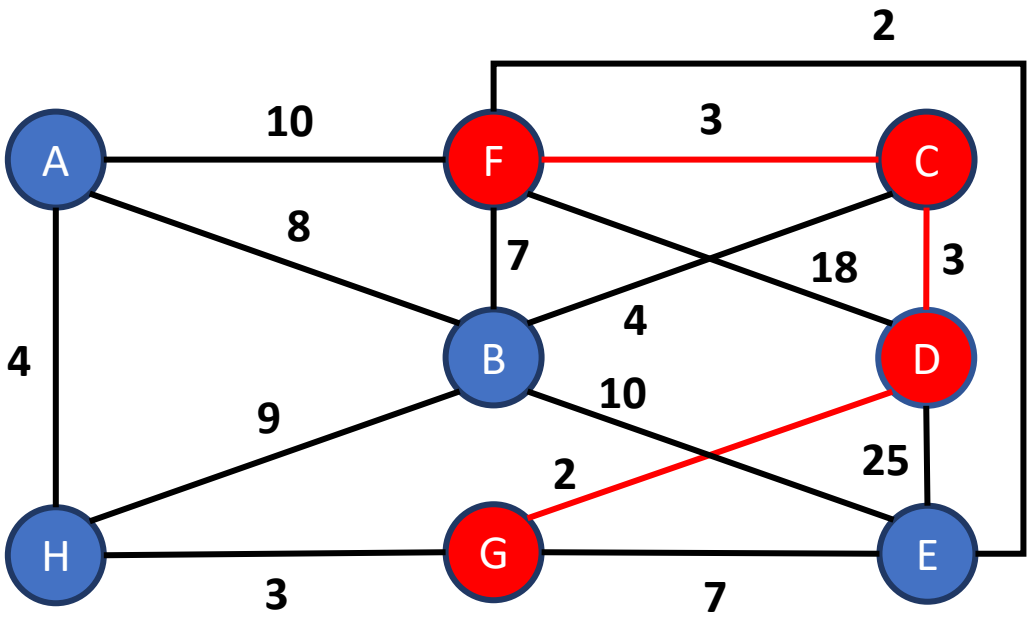



| $v$ | Selected | $d_v$ | $\pi_v$ |
|---|---|---|---|
| A | | | |
| B | | 4 | C |
| C | T | 3 | D |
| D | T | 0 | - |
| E | | ~~25~~ 7 | G |
| F | | ~~18~~ 3 | ~~D~~ C |
| G | T | 2 | D |
| H | | 3 | G |

Update distances of adjacent, unselected vertices

Sensitivity: Internal

# Prim's Algorithm

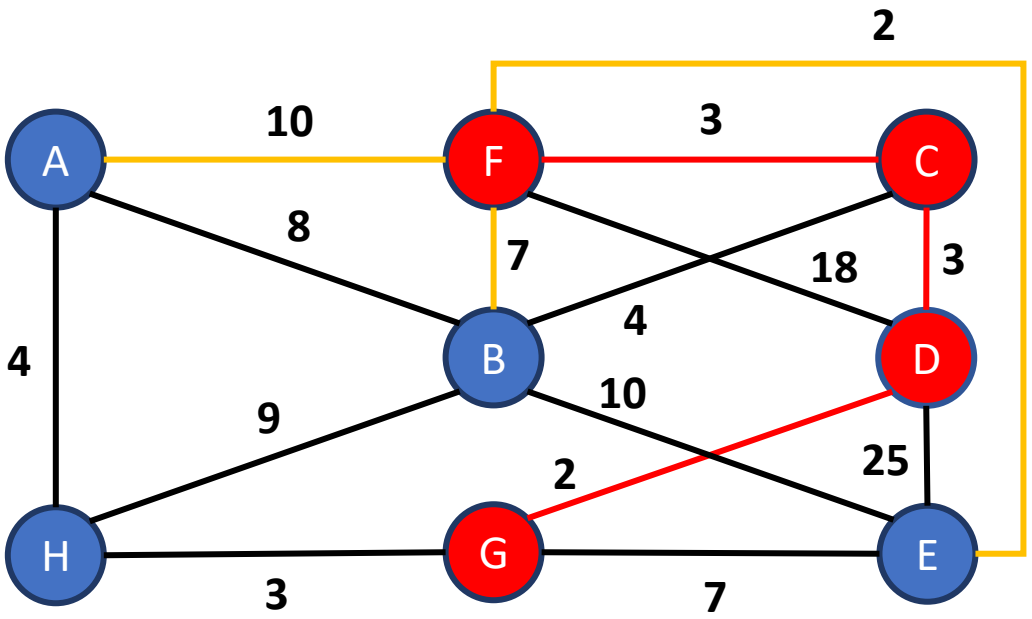



| $v$ | Selected | $d_v$ | $\pi_v$ |
|---|---|---|---|
| A | | | |
| B | | 4 | C |
| C | T | 3 | D |
| D | T | 0 | - |
| E | | ~~25~~ 7 | G |
| F | T | ~~18~~ 3 | ~~D~~ C |
| G | T | 2 | D |
| H | | 3 | G |

Select reachable vertex with minimum distance

Sensitivity: Internal

# Prim's Algorithm

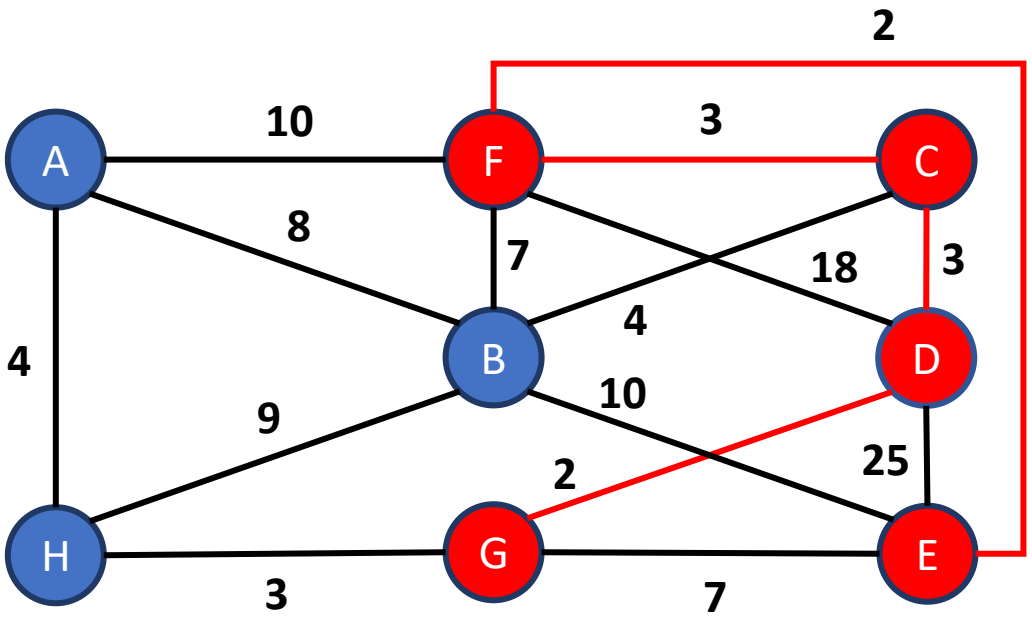



| $v$ | Selected | $d_v$ | $\pi_v$ |
|---|---|---|---|
| A | | 10 | F |
| B | | 4 | C |
| C | T | 3 | D |
| D | T | 0 | - |
| E | | ~~25~~ ~~7~~ 2 | ~~G~~ F |
| F | T | ~~18~~ 3 | ~~D~~ C |
| G | T | 2 | D |
| H | | 3 | G |

Update distances of adjacent, unselected vertices

Sensitivity: Internal

# Prim's Algorithm

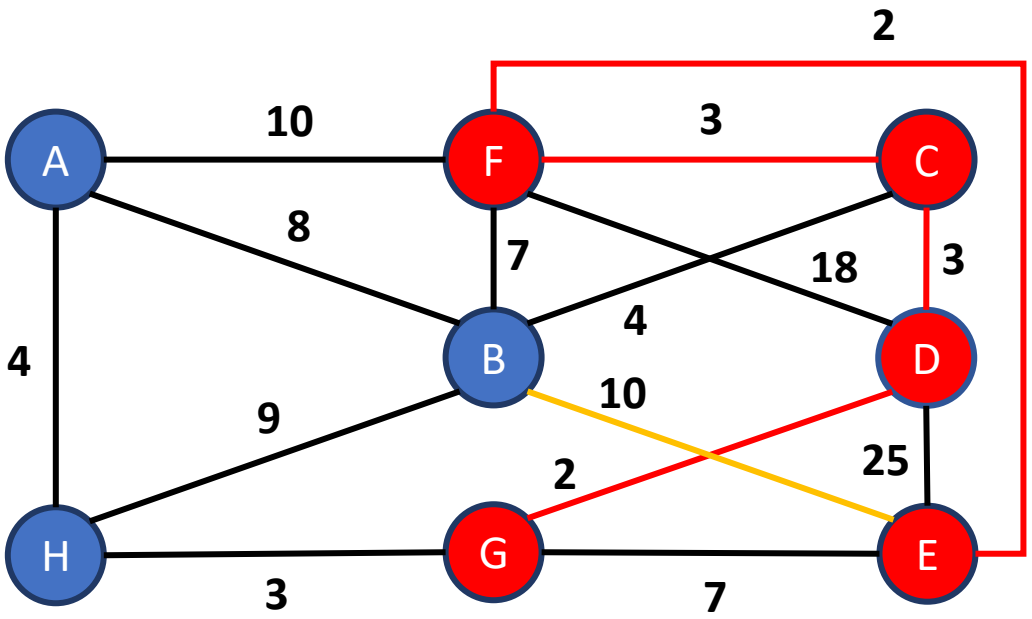



| $v$ | Selected | $d_v$ | $\pi_v$ |
|---|---|---|---|
| A | | 10 | F |
| B | | 4 | C |
| C | T | 3 | D |
| D | T | 0 | - |
| E | T | ~~25~~ ~~7~~ 2 | ~~G~~ F |
| F | T | ~~18~~ 3 | ~~D~~ C |
| G | T | 2 | D |
| H | | 3 | G |

Select reachable vertex with minimum distance

Sensitivity: Internal

# Prim's Algorithm

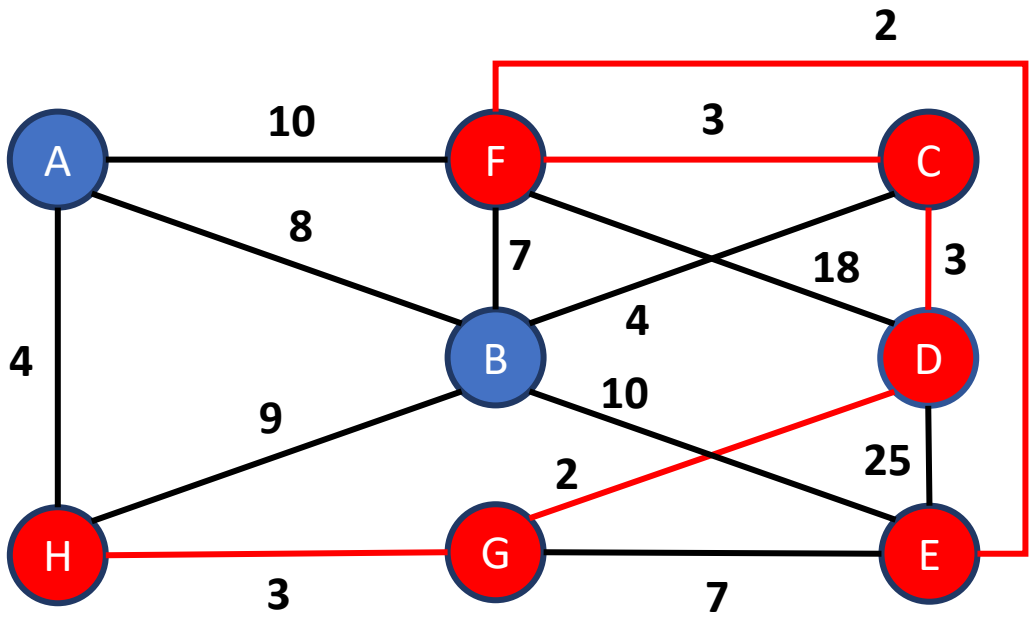



| $v$ | Selected | $d_v$ | $\pi_v$ |
|---|---|---|---|
| A | | 10 | F |
| B | | 4 | C |
| C | T | 3 | D |
| D | T | 0 | - |
| E | T | ~~25~~ ~~7~~ 2 | ~~G~~ F |
| F | T | ~~18~~ 3 | ~~D~~ C |
| G | T | 2 | D |
| H | | 3 | G |

Update distances of adjacent, unselected vertices

Sensitivity: Internal

# Prim's Algorithm

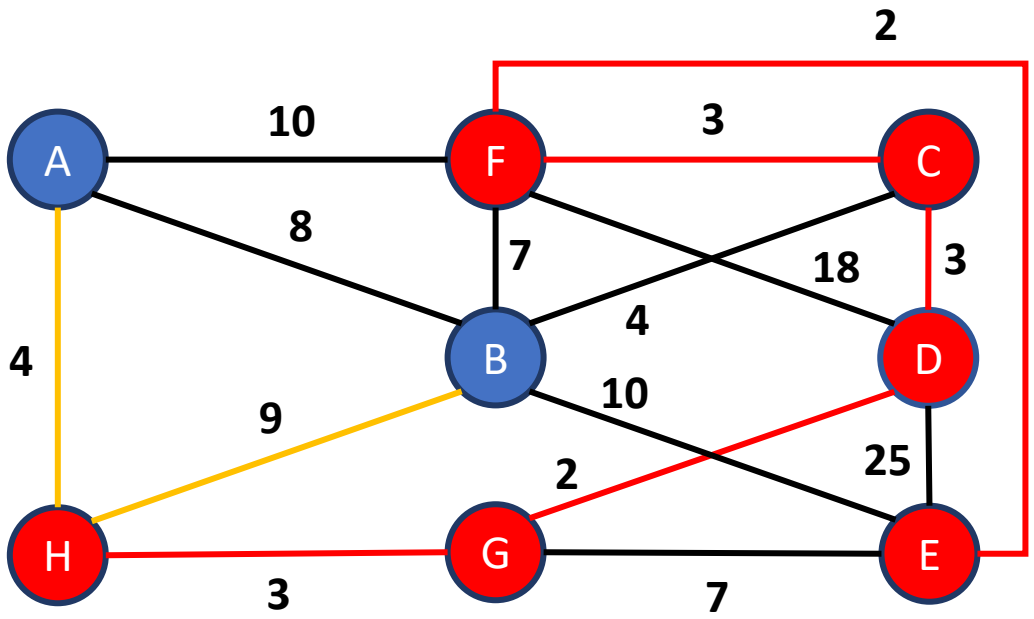



| $v$ | Selected | $d_v$ | $\pi_v$ |
|---|---|---|---|
| A | | 10 | F |
| B | | 4 | C |
| C | T | 3 | D |
| D | T | 0 | - |
| E | T | ~~25~~ ~~7~~ 2 | ~~G~~ F |
| F | T | ~~18~~ 3 | ~~D~~ C |
| G | T | 2 | D |
| H | T | 3 | G |

Select reachable vertex with minimum distance

Sensitivity: Internal

# Prim's Algorithm

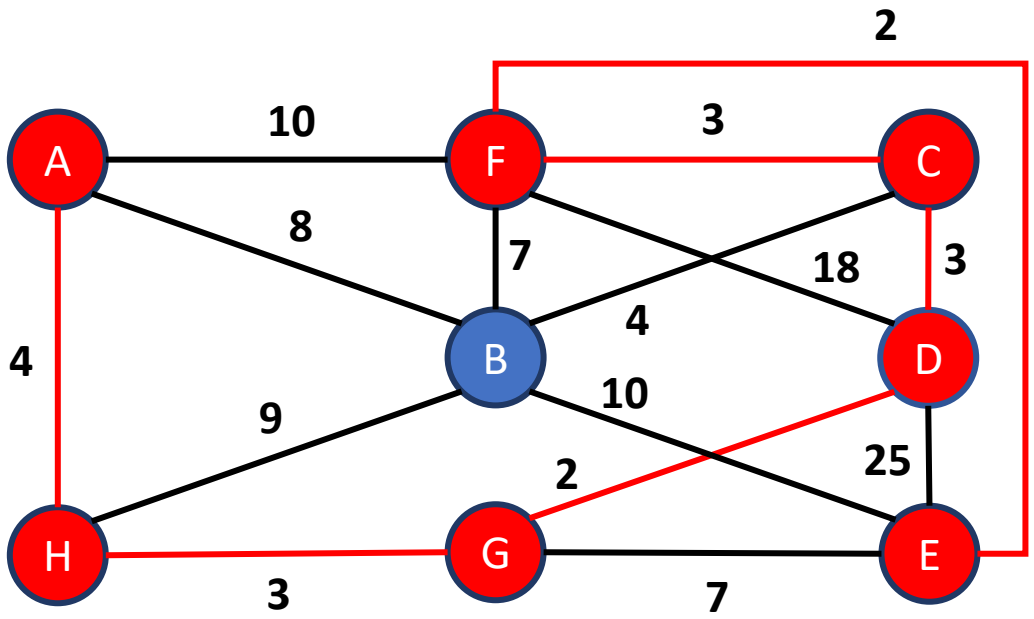



| $v$ | Selected | $d_v$ | $\pi_v$ |
|---|---|---|---|
| A | | ~~10~~ 4 | ~~F~~ H |
| B | | 4 | C |
| C | T | 3 | D |
| D | T | 0 | - |
| E | T | ~~25~~ ~~7~~ 2 | ~~G~~ F |
| F | T | ~~18~~ 3 | ~~D~~ C |
| G | T | 2 | D |
| H | T | 3 | G |

Update distances of adjacent, unselected vertices

Sensitivity: Internal

# Prim's Algorithm

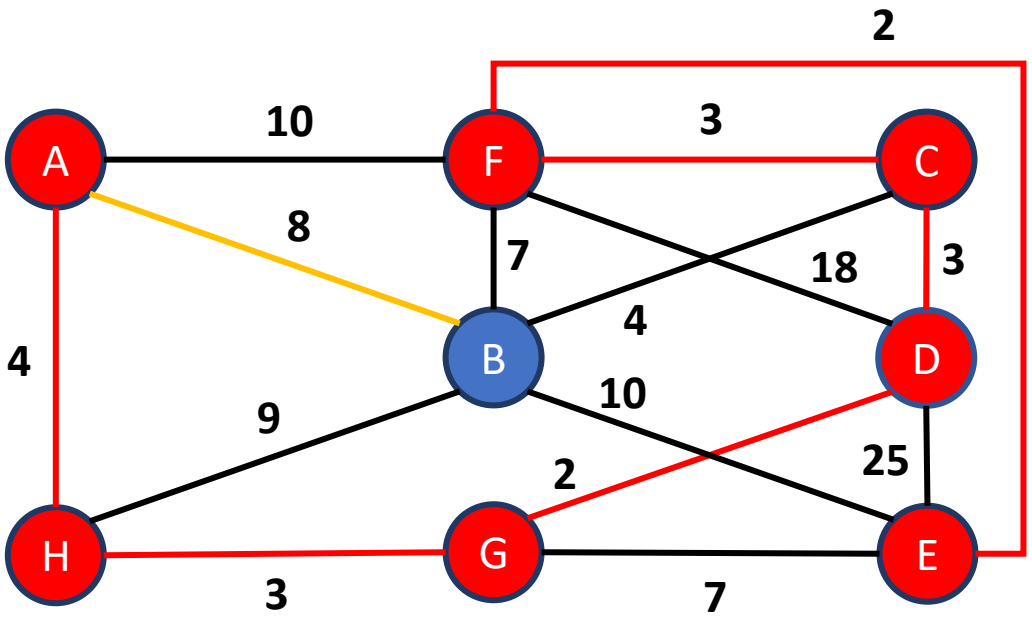



| $v$ | Selected | $d_v$ | $\pi_v$ |
|---|---|---|---|
| A | T | ~~10~~ 4 | ~~F~~ H |
| B | | 4 | C |
| C | T | 3 | D |
| D | T | 0 | - |
| E | T | ~~25~~ ~~7~~ 2 | ~~G~~ F |
| F | T | ~~18~~ 3 | ~~D~~ C |
| G | T | 2 | D |
| H | T | 3 | G |

Select reachable vertex with minimum distance

Sensitivity: Internal

# Prim's Algorithm

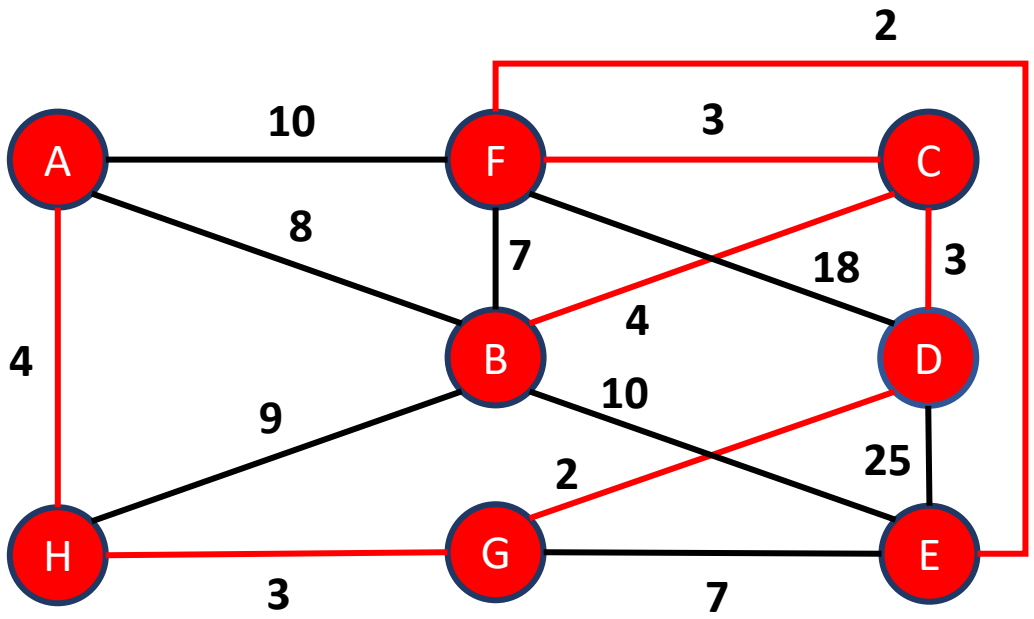



| $v$ | Selected | $d_v$ | $\pi_v$ |
|---|---|---|---|
| A | T | ~~10~~ 4 | ~~F~~ H |
| B | | 4 | C |
| C | T | 3 | D |
| D | T | 0 | - |
| E | T | ~~25~~ ~~7~~ 2 | ~~G~~ F |
| F | T | ~~18~~ 3 | ~~D~~ C |
| G | T | 2 | D |
| H | T | 3 | G |

Update distances of adjacent, unselected vertices

Sensitivity: Internal

# Prim's Algorithm

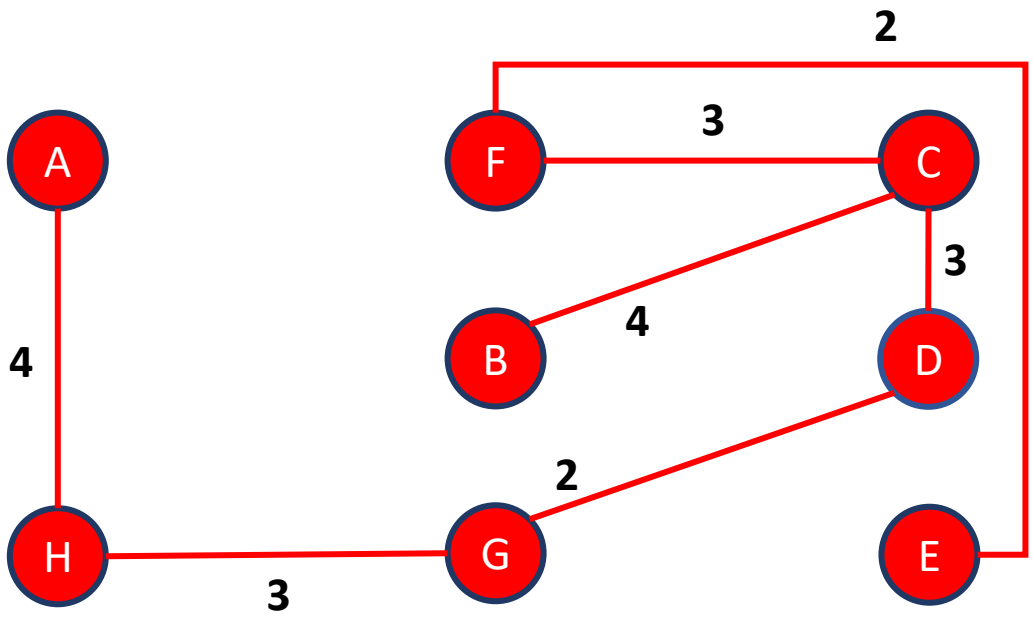



| $v$ | Selected | $d_v$ | $\pi_v$ |
|---|---|---|---|
| A | T | ~~10~~ 4 | ~~F~~ H |
| B | T | 4 | C |
| C | T | 3 | D |
| D | T | 0 | - |
| E | T | ~~25~~ ~~7~~ 2 | ~~G~~ F |
| F | T | ~~18~~ 3 | ~~D~~ C |
| G | T | 2 | D |
| H | T | 3 | G |

Select reachable vertex with minimum distance

Sensitivity: Internal

# Prim's Algorithm



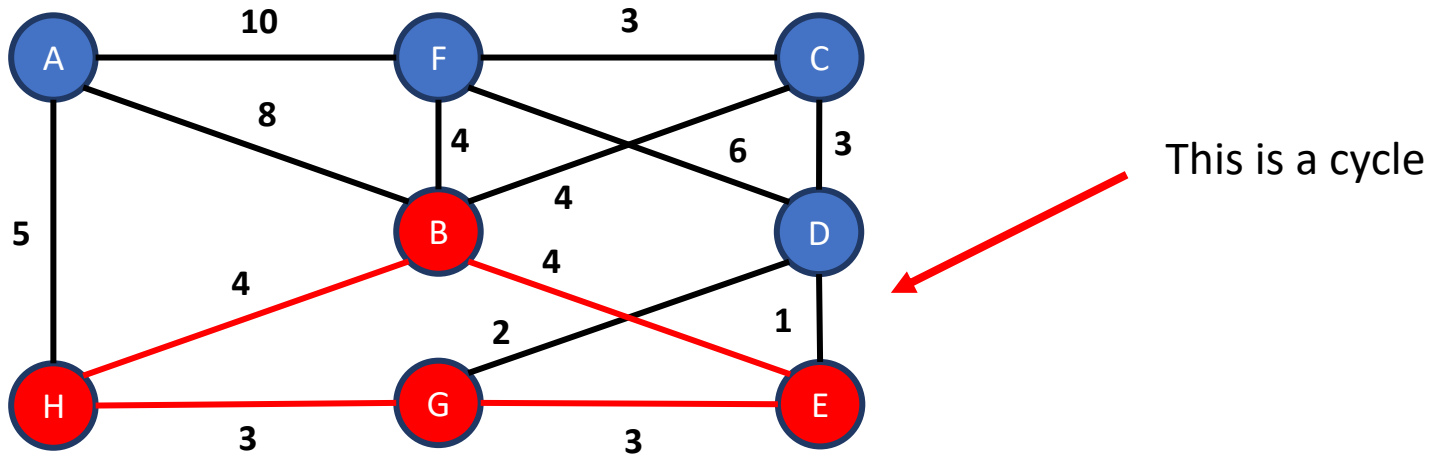

| $v$ | Selected | $d_v$ | $\pi_v$ |
|---|---|---|---|
| A | T | ~~10~~ 4 | ~~F~~ H |
| B | T | 4 | C |
| C | T | 3 | D |
| D | T | 0 | - |
| E | T | ~~25~~ ~~7~~ 2 | ~~G~~ F |
| F | T | ~~18~~ 3 | ~~D~~ C |
| G | T | 2 | D |
| H | T | 3 | G |

All vertices added and we have the minimum spanning tree. $\sum_v d_v = 21$

Sensitivity: Internal

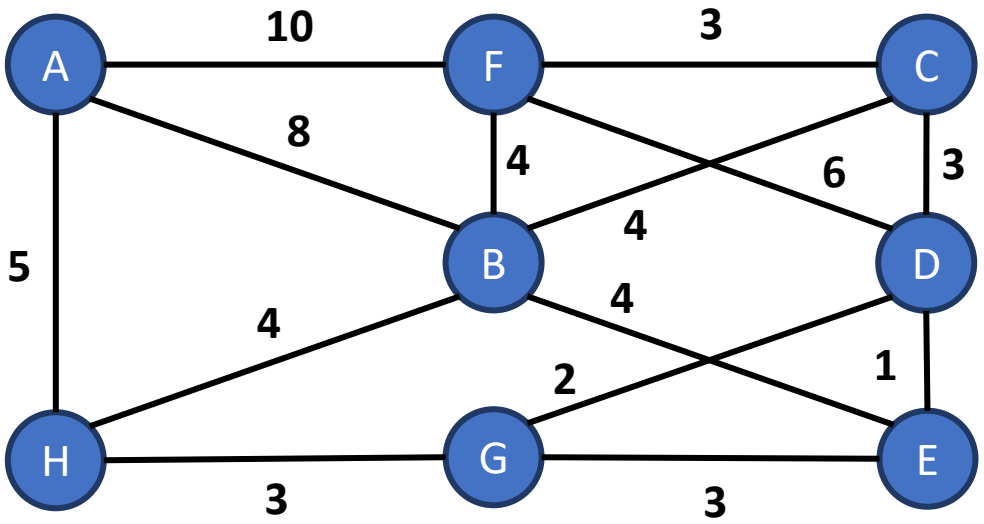
# Kruskal's Algorithm

1. Order the edges by weightings, lowest to highest
2. Select the lowest-weight edge not yet considered; if it does not form a cycle, label it as selected
3. Repeat step 2 until all vertices have been selected




Sensitivity: Internal

# Kruskal's Algorithm

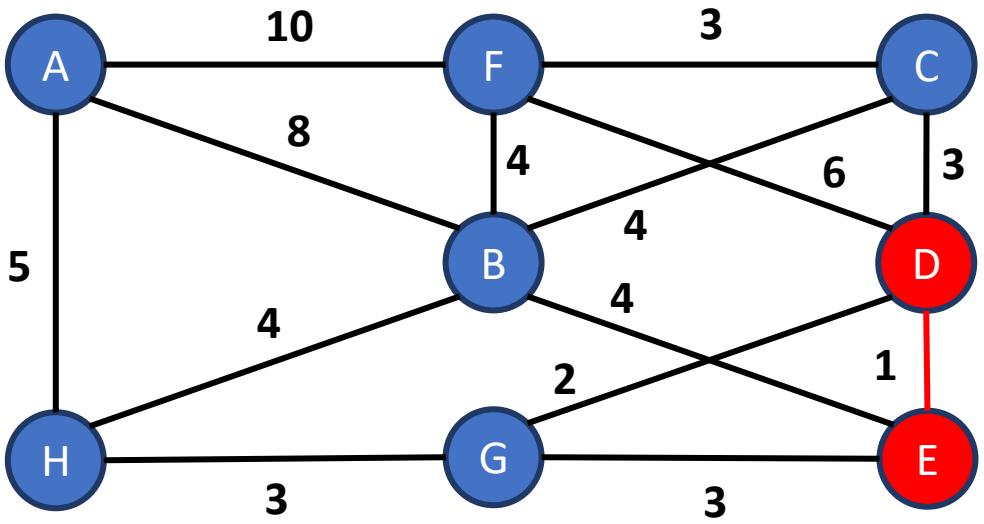



Select first $|V| - 1$ edges that do not form a cycle

| Edge $e$ | $w_e$ | Selected |
|---|---|---|
| (D,E) | 1 | |
| (D,G) | 2 | |
| (E,G) | 3 | |
| (C,D) | 3 | |
| (G,H) | 3 | |
| (C,F) | 3 | |
| (B,C) | 4 | |
| (B,E) | 4 | |
| (B,F) | 4 | |
| (B,H) | 4 | |
| (A,H) | 5 | |
| (D,F) | 6 | |
| (A,B) | 8 | |
| (A,F) | 10 | |

Sensitivity: Internal

# Kruskal's Algorithm

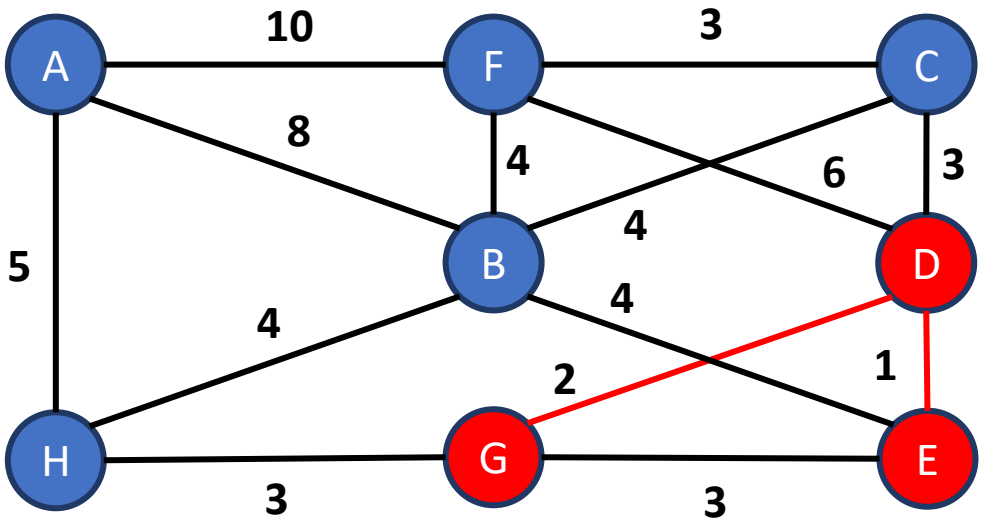



Select first $|V| - 1$ edges that do not form a cycle

| Edge $e$ | $w_e$ | Selected |
|---|---|---|
| (D,E) | 1 | T |
| (D,G) | 2 | |
| (E,G) | 3 | |
| (C,D) | 3 | |
| (G,H) | 3 | |
| (C,F) | 3 | |
| (B,C) | 4 | |
| (B,E) | 4 | |
| (B,F) | 4 | |
| (B,H) | 4 | |
| (A,H) | 5 | |
| (D,F) | 6 | |
| (A,B) | 8 | |
| (A,F) | 10 | |

Sensitivity: Internal

# Kruskal's Algorithm

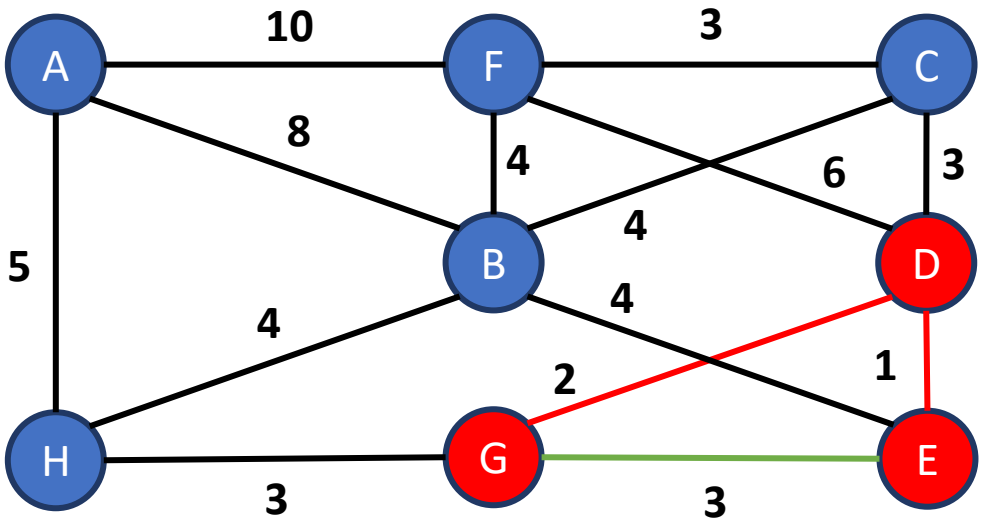



Select first $|V| - 1$ edges that do not form a cycle

| Edge | $w_e$ | Selected |
|---|---|---|
| (D,E) | 1 | T |
| (D,G) | 2 | T |
| (E,G) | 3 | |
| (C,D) | 3 | |
| (G,H) | 3 | |
| (C,F) | 3 | |
| (B,C) | 4 | |
| (B,E) | 4 | |
| (B,F) | 4 | |
| (B,H) | 4 | |
| (A,H) | 5 | |
| (D,F) | 6 | |
| (A,B) | 8 | |
| (A,F) | 10 | |

Sensitivity: Internal

# Kruskal's Algorithm



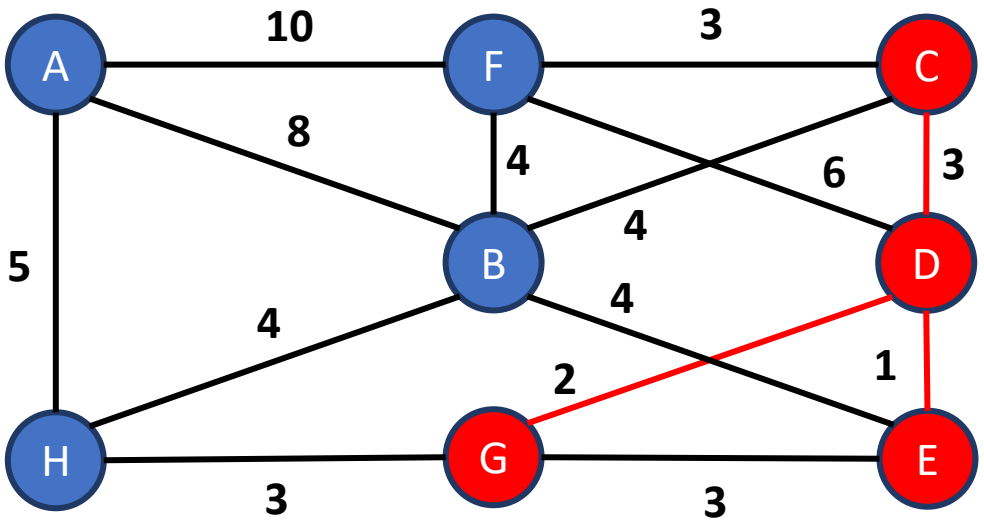

Forms a cycle!

Select first $|V| - 1$ edges that do not form a cycle

| Edge $e$ | $w_e$ | Selected |
|---|---|---|
| (D,E) | 1 | T |
| (D,G) | 2 | T |
| (E,G) | 3 | F |
| (C,D) | 3 | |
| (G,H) | 3 | |
| (C,F) | 3 | |
| (B,C) | 4 | |
| (B,E) | 4 | |
| (B,F) | 4 | |
| (B,H) | 4 | |
| (A,H) | 5 | |
| (D,F) | 6 | |
| (A,B) | 8 | |
| (A,F) | 10 | |

Sensitivity: Internal

# Kruskal's Algorithm

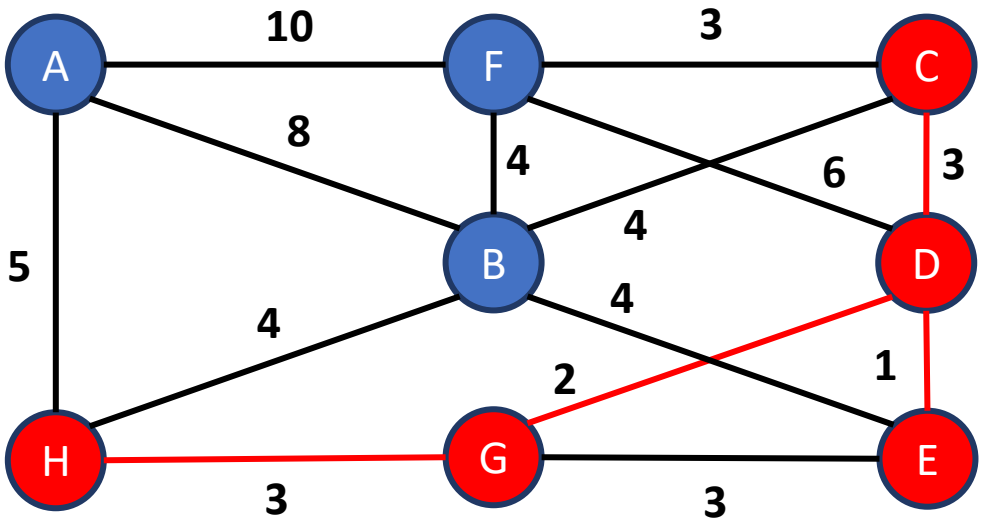



Select first $|V| - 1$ edges that do not form a cycle

| Edge $e$ | $w_e$ | Selected |
|---|---|---|
| (D,E) | 1 | T |
| (D,G) | 2 | T |
| (E,G) | 3 | F |
| (C,D) | 3 | T |
| (G,H) | 3 | |
| (C,F) | 3 | |
| (B,C) | 4 | |
| (B,E) | 4 | |
| (B,F) | 4 | |
| (B,H) | 4 | |
| (A,H) | 5 | |
| (D,F) | 6 | |
| (A,B) | 8 | |
| (A,F) | 10 | |

Sensitivity: Internal

# Kruskal's Algorithm

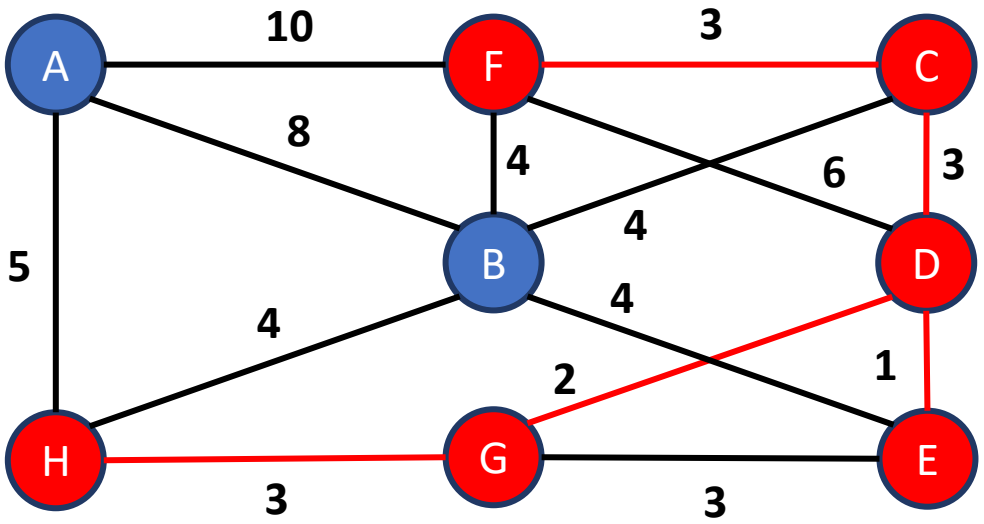



Select first $|V| - 1$ edges that do not form a cycle

| Edge $e$ | $w_e$ | Selected |
|---|---|---|
| (D,E) | 1 | T |
| (D,G) | 2 | T |
| (E,G) | 3 | F |
| (C,D) | 3 | T |
| (G,H) | 3 | T |
| (C,F) | 3 | |
| (B,C) | 4 | |
| (B,E) | 4 | |
| (B,F) | 4 | |
| (B,H) | 4 | |
| (A,H) | 5 | |
| (D,F) | 6 | |
| (A,B) | 8 | |
| (A,F) | 10 | |

Sensitivity: Internal

# Kruskal's Algorithm

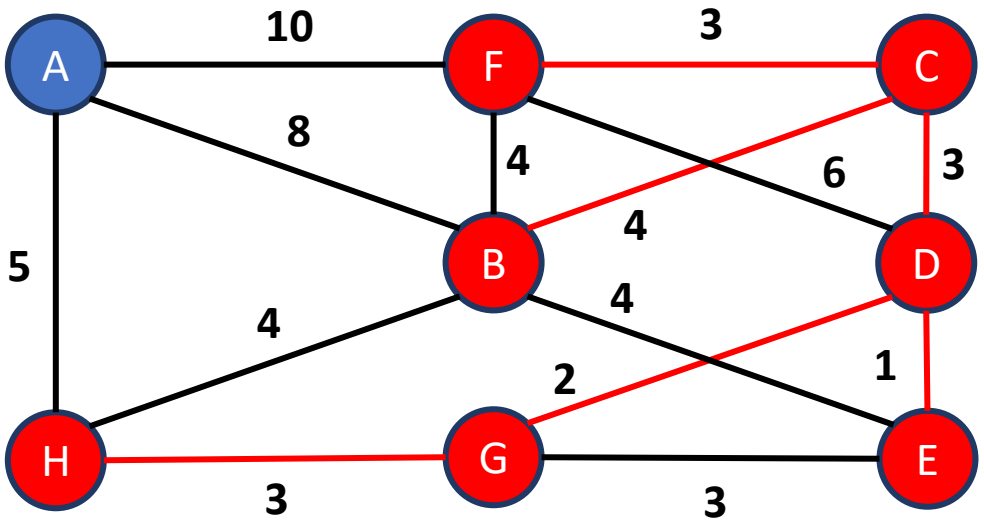



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| (C,D) | 3 | T |
| (G,H) | 3 | T |
| (C,F) | 3 | T |
| (B,C) | 4 | |
| (B,E) | 4 | |
| (B,F) | 4 | |
| (B,H) | 4 | |
| (A,H) | 5 | |
| (D,F) | 6 | |
| (A,B) | 8 | |
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Sensitivity: Internal

# Kruskal's Algorithm

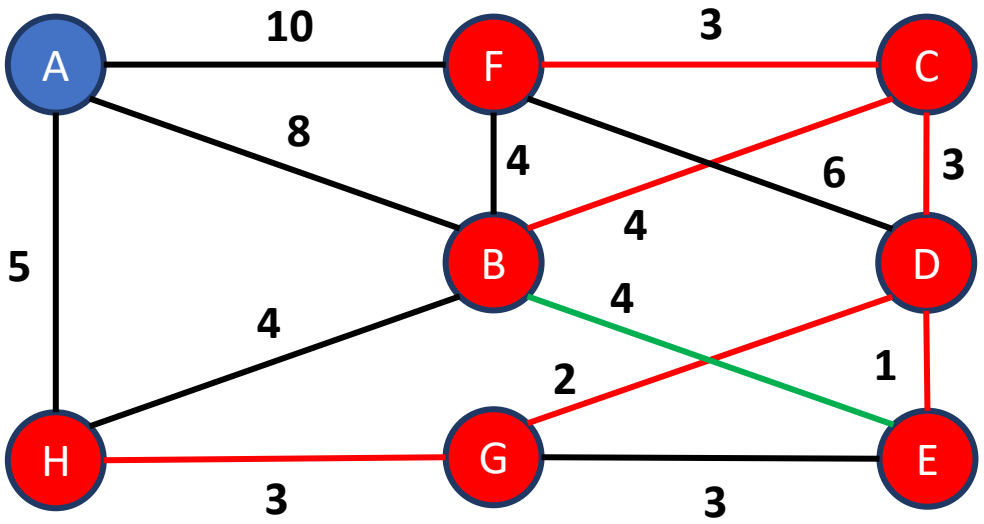



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| (E,G) | 3 | F |
| (C,D) | 3 | T |
| (G,H) | 3 | T |
| (C,F) | 3 | T |
| (B,C) | 4 | T |
| (B,E) | 4 | |
| (B,F) | 4 | |
| (B,H) | 4 | |
| (A,H) | 5 | |
| (D,F) | 6 | |
| (A,B) | 8 | |
| (A,F) | 10 | |

Sensitivity: Internal

# Kruskal's Algorithm



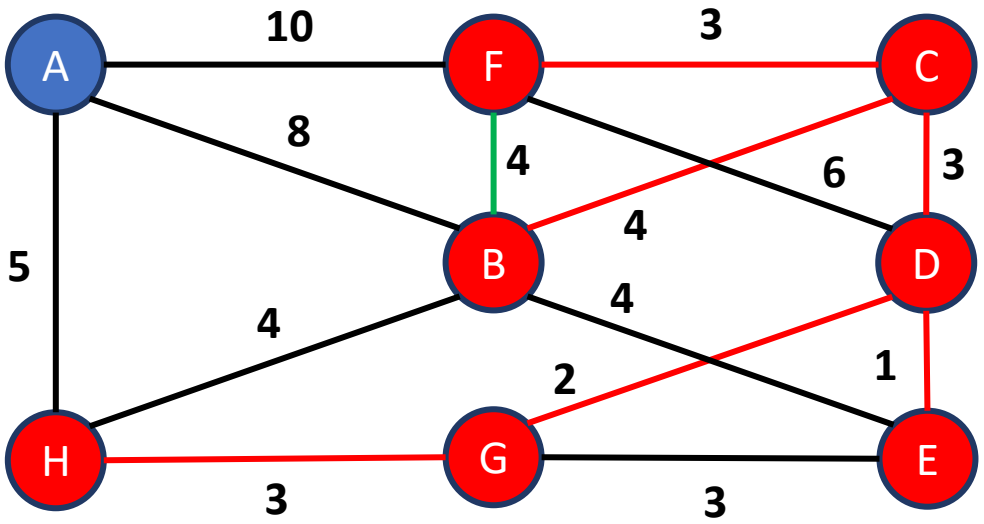

Forms a cycle!

Select first $|V| - 1$ edges that do not form a cycle

| Edge $e$ | $w_e$ | Selected |
|---|---|---|
| (D,E) | 1 | T |
| (D,G) | 2 | T |
| (E,G) | 3 | F |
| (C,D) | 3 | T |
| (G,H) | 3 | T |
| (C,F) | 3 | T |
| (B,C) | 4 | T |
| (B,E) | 4 | F |
| (B,F) | 4 | |
| (B,H) | 4 | |
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| (D,F) | 6 | |
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Sensitivity: Internal

# Kruskal's Algorithm



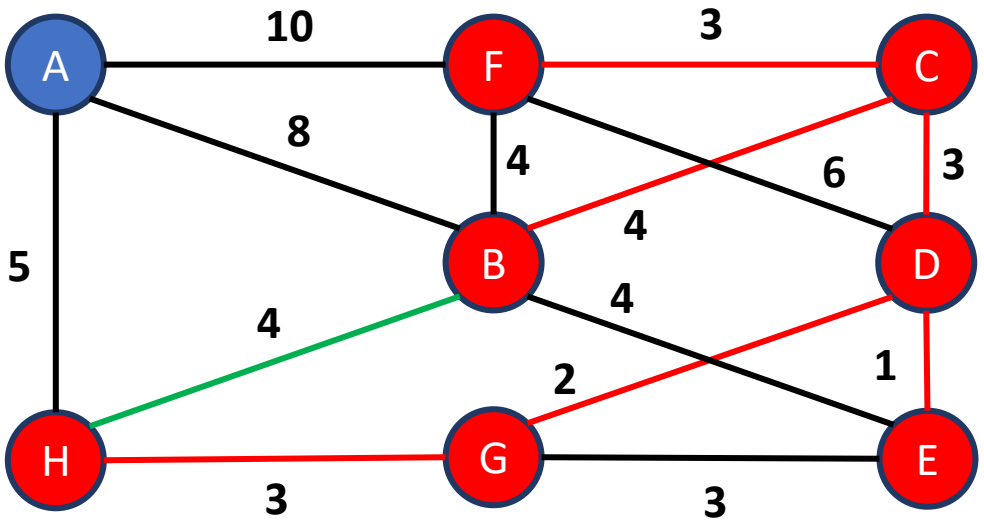

Forms a cycle!

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| (C,D) | 3 | T |
| (G,H) | 3 | T |
| (C,F) | 3 | T |
| (B,C) | 4 | T |
| (B,E) | 4 | F |
| (B,F) | 4 | F |
| (B,H) | 4 | |
| (A,H) | 5 | |
| (D,F) | 6 | |
| (A,B) | 8 | |
| (A,F) | 10 | |

Sensitivity: Internal

# Kruskal's Algorithm



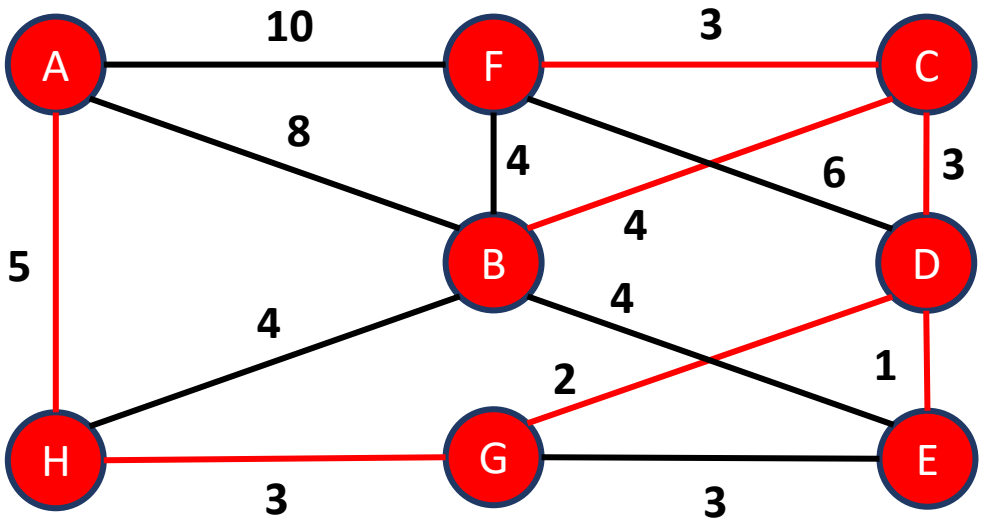

Forms a cycle!

Select first $|V| - 1$ edges that do not form a cycle

| Edge $e$ | $w_e$ | Selected |
|---|---|---|
| (D,E) | 1 | T |
| (D,G) | 2 | T |
| (E,G) | 3 | F |
| (C,D) | 3 | T |
| (G,H) | 3 | T |
| (C,F) | 3 | T |
| (B,C) | 4 | T |
| (B,E) | 4 | F |
| (B,F) | 4 | F |
| (B,H) | 4 | F |
| (A,H) | 5 | |
| (D,F) | 6 | |
| (A,B) | 8 | |
| (A,F) | 10 | |

Sensitivity: Internal

# Kruskal's Algorithm



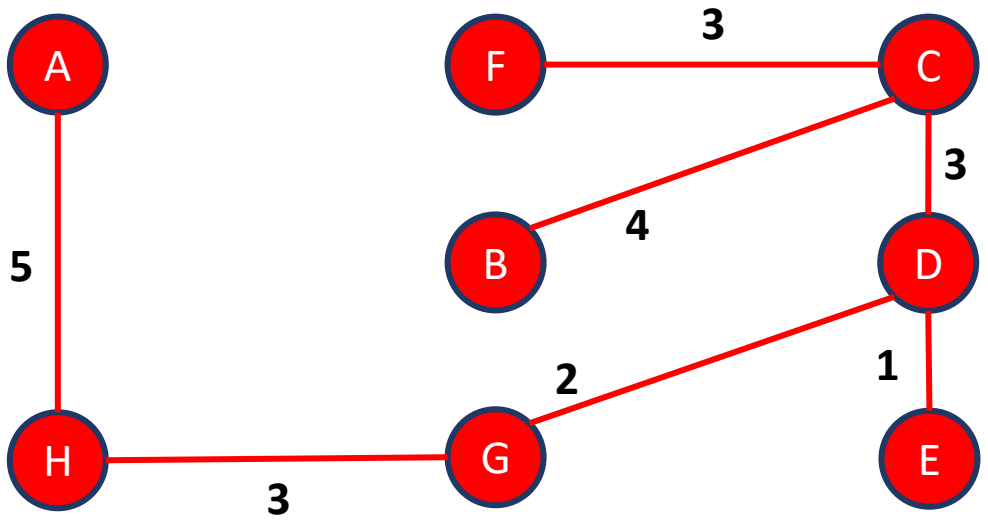

All vertices are selected. Stop

Select first $|V| - 1$ edges that do not form a cycle

| Edge $e$ | $w_e$ | Selected |
|---|---|---|
| (D,E) | 1 | T |
| (D,G) | 2 | T |
| (E,G) | 3 | F |
| (C,D) | 3 | T |
| (G,H) | 3 | T |
| (C,F) | 3 | T |
| (B,C) | 4 | T |
| (B,E) | 4 | F |
| (B,F) | 4 | F |
| (B,H) | 4 | F |
| (A,H) | 5 | T |
| (D,F) | 6 | |
| (A,B) | 8 | |
| (A,F) | 10 | |

Sensitivity: Internal

# Kruskal's Algorithm




| Edge $e$ | $w_e$ | Selected |
|---|---|---|
| (D,E) | 1 | T |
| (D,G) | 2 | T |
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| (B,C) | 4 | T |
| (B,E) | 4 | F |
| (B,F) | 4 | F |
| (B,H) | 4 | F |
| (A,H) | 5 | T |
| (D,F) | 6 | F |
| (A,B) | 8 | F |
| (A,F) | 10 | F |

$|V| - 1$ edges selected, let $A$ be the edges in the MST, then

$$\sum_{e \in A} w_e = 21$$

Sensitivity: Internal

# References

Cormen, T. H., Leiserson, C. E., Rivest, R. L., Stein, C. (2001). *Introduction to Algorithms*. The MIT Press. ISBN: 0262032937

Taha, H. A. (2017). Operations Research: An Introduction (10th ed.). Pearson.

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